

Mapping & Value Chain Analysis of the Fishery Sub-Sector in Somalia



TECHNICAL REPORT

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This technical report was conducted by UNIDO Somalia Programme team of experts, within the framework of the “Agro-technology development for Economic Growth in South and Central Somalia” project, implemented by UNIDO and funded by AICS/Italy.



Acronyms

1.	ODA	Overseas Development Assistance
2.	MSME	Micro Small & Medium Enterprises
3.	GDP	Gross Domestic Product
4.	FAO	Food & Agriculture Organization
5.	EEZ	Exclusive Economic Zone
6.	IUU	Illegal Unreported Unregulated
7.	IOTC	Indian Ocean Tuna Commission
8.	IUCN	International Union for Conservation of Nature
9.	UNEP	United Nations Environment Programme
10.	MT	Metric Tonne
11.	MFMR	Ministry of Fisheries and Marine Resources
12.	VOA	Voice of America
13.	ICRC	International Committee of Red Cross
14.	NGO	Non-Governmental Organization
15.	OEC	Observatory of Economic Complexity
16.	IORA	Indian Ocean Rim Association
17.	FSNAU	Food Security & Nutrition Analysis Unit
18.	SATG	Somali Agricultural Technical Group

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1. INTRODUCTION

1.1. Project Background

The Agro-Tech project which is being implemented by UNIDO is providing assistance in the revival of productive sectors in Central and South Somalia regions by establishing a network of three agro-technology and enterprise development units to support Somali businesses in starting, rehabilitating and upgrading agro-industrial operations, particularly by accessing to new technologies, markets, market-based skilled labour and financing facilities.

Emphasis in the project implementation will also be provided in supporting the labour force responding to the needs of the market with market based vocational and technical skills trainings to feed the growing needs of the MSMEs sector with skilled labour capable of making modern and better tools, and in installing, operating, maintaining and repairing old and modern agriculture and agro-industrial equipment.

Furthermore, a credit facility to facilitate access to finance to project supported entrepreneurs is to be established and deployed within the framework of the project.

Overall, the project will aim at increasing the potential for economic opportunities and job creation in the agro-industrial sector of the Southern and Central regions of Somalia by focusing on four key actions:

- i. Promoting agro-technology upgrading and entrepreneurship development;
- ii. Delivering vocational and technical skill trainings to improve productivity of agro-industrial tradesmen and micro-industries
- iii. Facilitating access to technology, markets and finance; and
- iv. Providing institutional support in the area of agro-technology transfer, investment promotion, and entrepreneurship development.

In order to design the vocational training curriculum, the modules will be made based on skill gap analysis in three of the intervention sub-sectors- Fruits & Vegetables, Livestock and Fishery. Hence, to have an understanding of these three sub-sectors, a sub-sector mapping and value chain analysis is planned for each of these three sub-sectors and provide recommendation in the areas of the value chains where the skill gap analysis could look at.

This project has been funded by the Italian Cooperazione with FGS- Ministry of Commerce & Industry (MoCI) as the line Ministry.

1.2. Socio-economic profile of Somalia

Facing decades' civil war, Somalia has gradually stabilized its economy with institutions and systems being built. The country is still fragile with highly dependent on aid and remittances, which is reflected when drought or major economic upheavals occur. As per the World Bank report the GDP of the country is growing at a steady rate (2 to 3 %) since 2013 with an exception of 2017 when due to a slump, the GDP shrunk. The 2018, nominal GDP for Somalia is **4.72 billion USD** with per capita GDP is 332 USD, which is not even 1 USD a day. The GDP estimate in Somalia is calculated using the expenditure method.

It can be seen that the substantial contribution to GDP is house-hold consumption and not government expenditure or industrial production. The negative net exports shown above shows that the Imports is

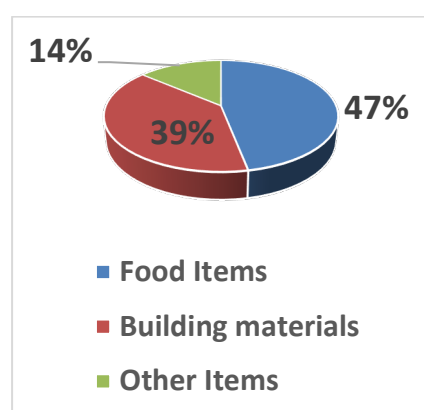
almost the same of GDP, which **indicates that the economy is highly import driven and is the main source for fuelling the house-hold consumption.**

An analysis of the import/export figures received from Mogadishu Port Authority (Fig. for year 2019). The tables below provide a snapshot on import items.

Item	Amount
Household final consumption expenditure	7,358
Government final consumption expenditure	531
Gross fixed capital formation • Private investment spending • Public investment spending	309
Net exports of goods and services	-3,478
	Export: 1,224
	Import: 4,702

All figures in million USD

An analysis of the import/export figures received from Mogadishu Port Authority (Fig. for year 2019). The tables below provide a snapshot on import items.



Food Items	Building Mat	Other Items
• Sugar • Rice • Flour • Pasta • Cooking Oil • Oatmeal • Milk powder • Tomato paste	• Cement • Wood/Iron • Iron sheet • Tiles /plates • Painting	• Garments • Electronics • Spare-parts • Drugs • Sanitary items

It can be seen from the above table that food import constitutes a significant amount which is further substantiated by FAO-World Bank report that states Somalia faces chronic food crop deficit with own production can only 22% of the per capita needs.

The limited exports that goes from Somalia is highly agriculture oriented with 93% share is from Agriculture of which Livestock (Whole animal) presents 50% of the total. The other agriculture items are Sesame and dry lemon which is exported. Most of the live animal is exported to Saudi Arabia- specially catering to Hajj visitors and during Eid Al-Adha.

While, official remittances from the diaspora are estimated at about \$1.4 billion a year, equivalent to about 23 percent of Somalia's GDP (IMF 2017) and could be the major contributor to house-hold private consumption that is the back-bone of Somalia's GDP. About 3.4 million people—roughly 40 percent of the population—depend on remittances for their daily needs, and about 80 percent of all new business ventures are funded by remittances. The IMF figures for 2017 quotes that Somalia received foreign assistance as ODA is 1.65 billion USD.

Somalia, being import depended country with minimal exports, the balance of payment is highly skewed with trade deficit as mentioned in 2018 as 3478 million USD. This negative balance of payment is

normally covered through foreign remittances (almost 25% of GDP) and external grants classified as ODA (Overseas Development Assistance).

1.3.Scope of the Study

- Mapping of the existing Fishery sub-sector in Somalia.
- Analyse various value chains operating within the sub-sector.
 - Production system
 - Demand and Supply Analysis
 - Value addition within the Chains
 - Marketing Channels
 - Actors/Sub-actors/Stake-holders
 - Horizontal and Vertical Linkages/Relationship among the Actors
 - Governance
 - Inter-firm cooperation
- Recommendations in selected areas of the operating value chains on the probable areas of skill set building.

1.4.Methodology/ Approach of the study

The methodology adopted was with twin pronged approach for conducting the studying which includes primary information collection & analysis and secondary data analysis for achieving the overall objectives.

The assessment of the Fishery subsector and its value chains was conducted by UNIDO, using the following steps:

- Scoping visit to the field
- Secondary literature review
- Study tools development
- Field research
- Analysis and report preparation

Each of these activities is detailed below:

a. Scoping visit to the field

The first step taken by the research team was to acquaint themselves with the ground realities and understand the broad parameters of the sub-sector operating in the country. The team interacted with various experts, key informants of the sub-sector and prepared a preliminary report, which was shared with the PMU.

This scoping visit helped the research team to have a preliminary idea about the various players in the value chain. This was useful in designing the various questionnaires that were used to collect data from field.

b. Secondary Literature Review

Different secondary literature was collected and reviewed to develop a general idea on the sub-sector and its associated value chains and the study area. Extensive web search was done to locate secondary information that would be useful for analysis. The secondary literature was continuously referred throughout the primary data collection as well as during analysis and report writing period. In addition, inputs from expert opinions from various institutions have been collected and used in the study.

A list of prominent articles and documents that were referred to has been provided as part of the annex

c. Study Tools Development

As a part of the primary data collection, it was decided to have individual survey of the value chain actors. **The individual survey was conducted with fishermen, traders, processors.** Separate sets of questionnaire/ checklist were developed for each actor and used in the field research.

d. Field research

The field research component was conducted for a period of 30 days. During this period, individual interview with fishermen, individual interviews with traders, individual interviews with key informants including processors of fish were conducted by team consisting of local expert and UNIDO local staff.

The interaction with stakeholders focused upon tapping the knowledge base of the local stakeholders, the status of fishery production in the country and also assesses their capacities for involvement in the activity.

The interactions and visits were supported by analysis of the secondary data related to fishery production and trade, and also helped in forecast and estimation.

e. Analysis and report writing

All the data collected from individual interviews were then fed into MS Excel sheets from the hard copies of the questionnaire. These data were then used to generate tables, which were analyzed to find the situation and the status of the value chain. These tables are then incorporated into the report, and the combination of the data analyses of primary data and inferences from the secondary literature reviewed coupled with opinion from key informants were the basis for the report.

1.5.Scope & Limitation of the study

Some of the limitations faced by the study team during its course of work is reported below:

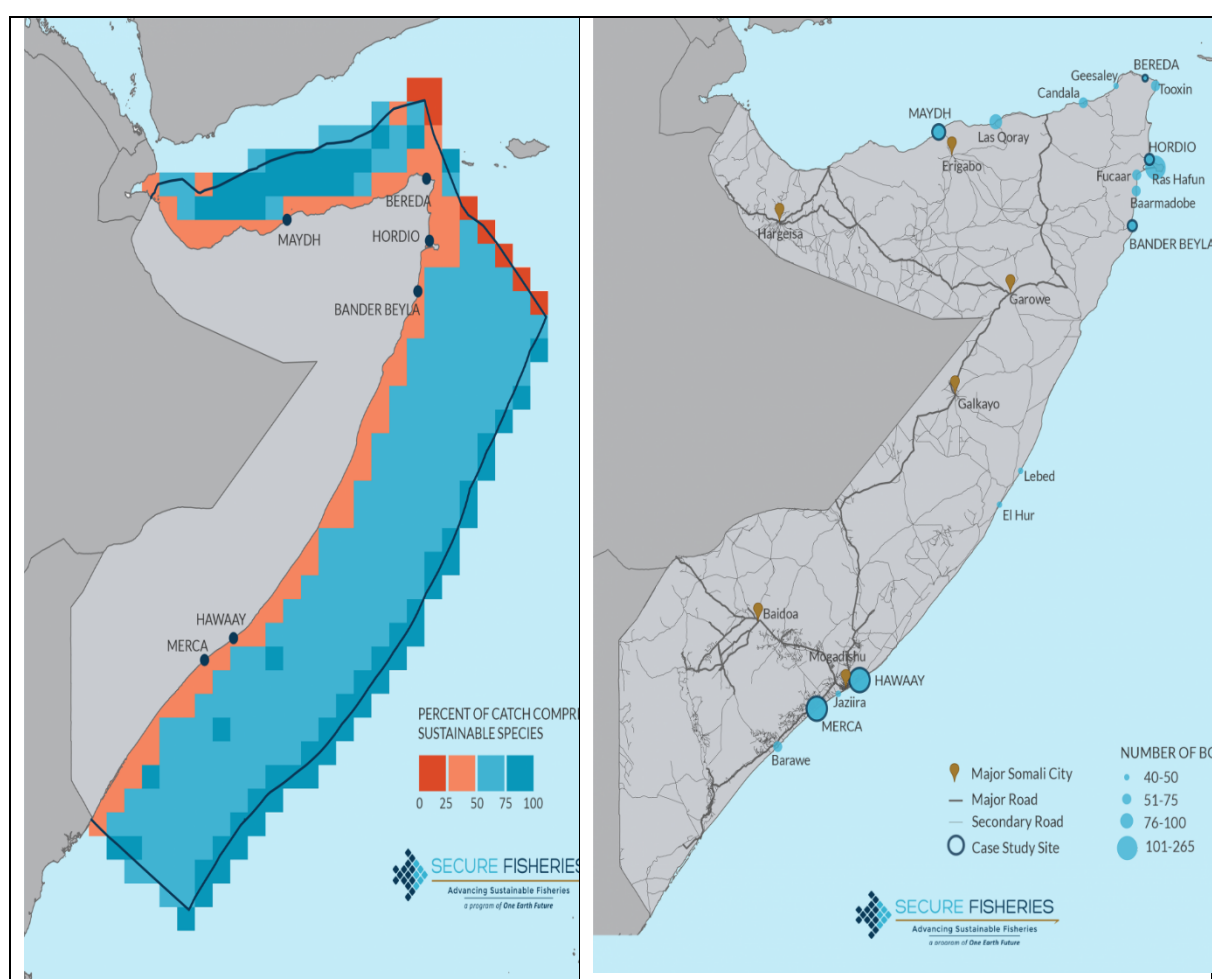
- The sample size for primary data has been selected mostly from Banadir and Juba-land, where accessibility of the research team was high. The input received from these samples along-with secondary data has been extrapolated to generalise for the country, which might have overlooked many local conditions prevalent in other regions.
- The primary data collection has been through re-call methodology, which has the limitation due to non-documentation of records as well as the fear of mixing of records pertaining to other crops.
- Artisanal Fishery production and trading is a highly un-organized economic activity, hence the authenticity of secondary data used is at times doubtful.
- Somalia has come out of civil war only few years back. Many of the research Institutions as well as Government entities for data collection are still in their infancy. Due to this situation, the availability of secondary data was very difficult and limited. Even, whatever secondary data reported, the authenticity is not beyond doubt.

2. OVERVIEW OF FISHERY SUB-SECTOR IN SOMALIA

2.1. Introduction to Fishery Sub-sector

Somalia with a 3300 km long coast-line should have been in the fore-front of fish exploitation in Africa. The country due to its long coast-line, has a large exclusive economic zone (EEZ) estimated at 825,052–831,000 square kilometres. Unfortunately for years of instability and civil war, the country has never developed its fishery sub-sector for contributing significantly to the country's GDP. According to FAO, the contribution of Fishery to the GDP ranges around 1%, which indicates how under developed the sub-sector is. The figure below provides a snapshot of the geographical areas under fishing activities in Somalia.

Figure 1 : Snapshot of the fishing zones in Somalia



Source: Project Badweyn: Secure Fisheries

The Somali marine ecosystem, due to its narrow coastal shelf, except of the Puntland coast, which makes limited bottom feeder fishes or the demersal types availability. The EEZ has presence of large pelagic (highly migratory) species (such as tuna) to smaller pelagic fish, crustaceans, and other reef species.

Commercial exploitation of fishing never took place primarily due to the fact that Somalis are traditionally pastoralist with meat being the major source of protein. Fishing activity has

been practised by only limited group of communities that live in the coastal areas. As per SATG group report, only about 0.3% of the population is dependent on fishing as livelihood option. With per capita consumption of 3.1 kg (data as of 2009), fish consumption in Somalia is the lowest amongst the all African nations. Negligible local market and years of civil war preventing investment in the fishery sector has kept the sub-sector under-developed.

2.2. Classification of Fishery in Somalia

2.2.1: Source of Fish

Fishery in Somalia can be classified according to the geographical area of the fishing activities. Primarily it can be categorized into three source:

Inland Fishery:

The In-land fishery in Somalia is mostly practised by the communities living near the banks of two main rivers of Somalia- Juba and Shabelle. In addition, some wet-lands within the country too see this practice. For generations, share-cropping communities in Somalia have fished the lower reaches of southern Somalia's two permanent rivers. The common species found are **Catfish, Tilapia, Mullidae** etc. These fishes are highly valued locally and consumed by the local populace since these areas has shortage of livestock due the tsetse fly infestation.



Coastal or Inshore Fishery:

Most of the Somalia's EEZ is oceanic, because the continental shelf along its coastline is relatively narrow. All the fishing activities within this narrow zone is called the In-shore fishery where most of the catches are of the demersal variety. Many of the reef-associated, demersal species found in the coastal areas are the **grunt, emperor, grouper, goatfish, snapper, jack, seabream, and lizardfish** families. These fish are generally found close to shore, making them easier to catch from small boats with artisanal gear like handlines and nets. Species that

frequent reefs and inshore habitats, such as **blacktip reef shark**, **thintail thresher shark**, and **hammerhead sharks** are other varieties found in this zone, which come to look for demersal and pelagic varieties.

Off-shore Fishery:

Fishing activities conducted beyond 24 nautical miles from the coast can be classified as off-shore fishing. Commercially important species of large pelagic types, including tuna and tuna-like species, billfish, and sharks, are found in the offshore areas. Species that are caught in off-shore are migratory in nature and seasonal influences are very high on the type of species that are caught.

2.2.2: Nature of Fishery

The classification of fishery in Somalia can be further categorized by the nature of fishing activity. Primarily, it is divided into two types; **Artisanal or traditional fishing** and **Commercial fishing**.

Artisanal or traditional fishing:

This is the fishing style involving the local traditional fishermen community, mostly residing in the coastal areas – from Puntland to Kisimayo. Most of the Somalia's artisanal fleet is made of small wooden boats, mainly built locally from old designs and generally beach launched. In recent years, the introduction of fibre-glass made boats can be observed amongst the fishing community. The artisanal fishermen target coastal species using traps, spear, and nets and normally remains within the coastal zone. The men-folk of these coastal fishing communities carries out the fishing activity and fish processing. Women are involved in support activities and occasionally in trading. The artisanal or the traditional fishing is still a subsistence activity with only livelihood source for the fishermen community.

Commercial Fishing:

The fishing activity when done to economically and commercially to exploit the fish resources available in the Somalian maritime area using mechanized fishing boats and equipment is categorized as Commercial fishery. Interestingly, due to the country's instability, most of the commercial fishery can be termed as IUU activity. IUU or Illegal- Un-reported- Un-regulated. These IUU activities are undertaken by many fishermen from neighbouring countries with powerful trawlers and fishing in the off-shore areas. Since 2012, the Ministry of Fishery, gave licenses to some foreign companies to fish in the off-shore areas. Recently, the government has given license to 31 Chinese vessels to fish in the Somalian Off-shore areas.

2.3.Species of Fish Available

Primarily, the fish resources in Somalian Inshore and Off-shore area can be grouped into following categories:

Large Pelagic Stocks: The large pelagic fish are mainly tunas and big mackerels such as yellowfin tuna, bigeye tuna, long-tail tuna, bonito, skipjack tuna and Spanish mackerel. The quantum of these species vary greatly in accordance to the season of the year. The availability peaks in November and March but drastically falls during the South-West monsoon. The primary season for Spanish mackerel is March–June, and for tunas it is October–November.

Small Pelagic Stocks The dominant species are the Indian oil sardine, rainbow sardine, scads, chub mackerel, and horse mackerel. Anchovies belonging to this stock group also is found but lesser in abundance.

Demersal Species: These stock of fishes found mostly in the coastal reef includes the following varieties; Snappers, Groupers, Grunts, Sea breams, Goat fish, lizard fish etc. Several hundred species are exploited of this group by artisanal fishermen. These species support a year round fishery as they show limited migration patterns.

Elasmobranchs such as sharks and rays represent 40% of the artisanal catches. The principal groups are hammerheads, blacktip sharks, thin tail sharks etc.

Large Crustaceans: Most of the common lobsters caught by the artisanal fishermen are the Spiny lobsters of the genus *Panulirus*.

Figure 2: Clock-wise- Yellow-fin Tuna, Spanish Mackerel, Snapper, Spiny lobster

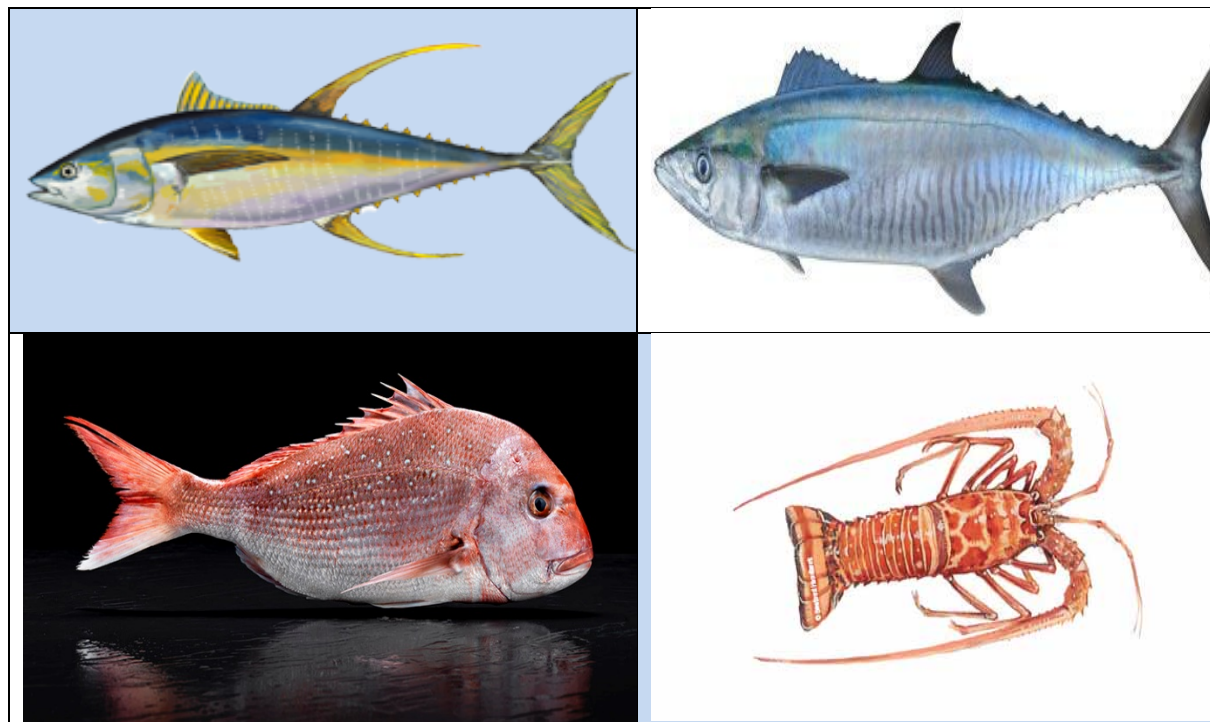


Figure 3: Economically important fishes & invertebrates found in Somali waters

		Category (in Somali)	Family	Common Size Range (cm)	Common Habitat	Catch Methods in Somali Waters
Coastal and Demersal Fishes		Emperors (gaxash, dhuuwane, miraamir, afdehere)	Lethrinidae	20-100	shallow to mid-depth, benthic, coastal, reefs, lagoons, seagrass	handline, trap, seine, gillnet, bottom trawl
		Goatfishes (gocoore, fangalaata, labogarile)	Mullidae	20-50	shallow to mid-depth, coastal, reefs, lagoons, sandy bottoms, muddy bottoms	pole and line, trap, beach seine, gillnet, bottom trawl
		Groupers (summaan, caalo, gadiir, yaagur)	Serranidae	14-270	shallow to mid-depth, demersal, reefs	pole and line, trap, bottom trawl
		Grunts (shoax, cadaasho)	Haemulidae	16-100	shallow, coastal, reefs	handline, trap, longline bottom trawl
		Jacks, Scads, Trevallies (bayaad, caari-joog, doorab, baroole, jabto)	Carangidae	18-170	shallow to mid-depth, coastal or pelagic, reefs	pole and line, handline, gillnet, trawl, purse seine, beach seine
		Snappers (booraad, qardabo, tartabo, cayje)	Lutjanidae	30-150	shallow to mid-depth, rocky bottoms, reefs	handline, longline, bottom trawl, gillnet
		Lizardfishes	Synodontidae	16-50	estuarine and marine, inshore	bottom trawl
		Seabream/ Porgies	Sparidae	30-90	shallow to mid-depth, benthic, coastal, reefs, muddy bottoms	bottom trawl, longline, handline, trap
Sharks and Rays		Sharks (geesoole, libaax, miishaar, seefle, saracilmo, nedaraan, baalcadde)	Sphyrnidae, Lamnidae, Alopiidae, Carcharhinidae	400-760	shallow and epipelagic, reefs and oceanic	longline, gillnet, handline
		Rays (safan, baalalay)	Dasyatidae, Myliobatidae	147-330	shallow, coastal, pelagic or benthic	longline, pole and line, handline, bottom trawl
Small Pelagic Fishes		Sardines, Herrings (carabi, buraasow, simbillig, caydi)	Clupeidae	7-24	pelagic, coastal, schooling	seine, gillnet, trawl, castnets
		Anchovies	Engraulidae	8-18	pelagic, coastal, schooling	seine, trawl, staknet, castnets
Large Pelagic Fishes		Billfishes (daabbiri, faras)	Istiophoridae	230-500	epipelagic, oceanic	longline, gillnet, troll
		Tuna (tabadin, yajdar-baal-cagaar, dhiglow ama shirwo, tabad, jeedar, sannuuro)	Scombridae	50-250	epipelagic, oceanic, some schooling	longline, gillnet, troll, handline, purse seine, pole and line
		Mackerel (derik, tabadin, jeedar)	Scombridae	38-240	epipelagic, coastal and oceanic, schooling	gillnet, trawl, purse seine, pole and line
		Dolphinfish	Coryphaenidae	75-210	pelagic, inshore and offshore	longline, purse seine, gillnet, pole and line
Invertebrates		Cuttlefish (sunaani, tootane)	Sepiidae	5-50	shallow to mid-depth, demersal	trawl, jig, trap
		Squid (duusho)	Loliginidae, Onychoteuthidae	30-100	shallow to deep, neritic	trawl, seine, jig
		Lobsters (arigoosto)	Palinuridae	9.5-50	shallow to deep, reefs, rocky, muddy, and sandy bottoms	trap, diving, trawl, spear, nets
		Shrimp and Prawns (ayax badeed, gaambiri, shiribis)	Penaeidae	7-34	shallow to mid-depth, benthic	trawl, seine, gillnet

Source: FAO.org

2.4. Estimated Catch & Potential

A detailed analysis of the estimated catch and the potential to tap has not yet been done thoroughly by any agencies. Different measuring tools were designed and with many regression models and forecast models were used to assess the estimates. The study report presents some inputs from various secondary sources to provide some insights on the species

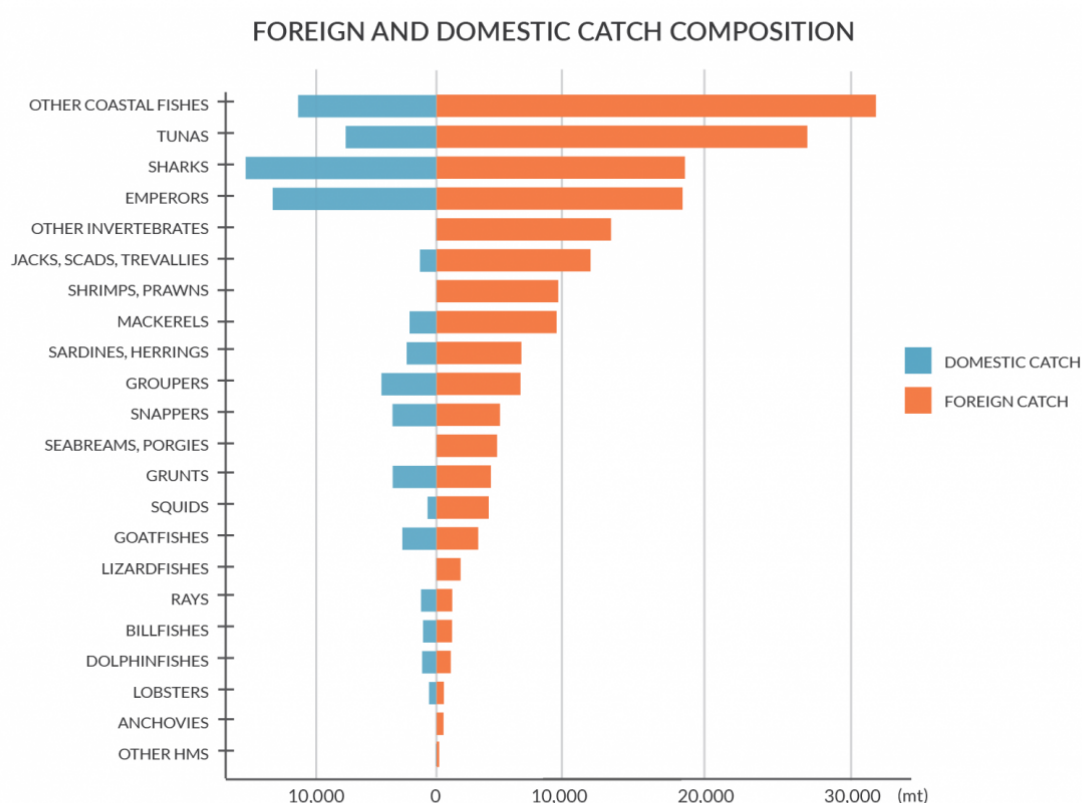
and the quantum of their landings. Since, a major component of off-shore fishing is through illegal commercial boats, the figures of their catch is always a large approximation.

In the chapter on Fish production system, catch inputs from the fishermen respondents have been reported but we are not still confident of extrapolating it for the whole country since the sample size is not truly reflective of the population.

Since, there is not a clear line of demarcation where demersel and pelagic stocks donot separate to be caught and similarly the artisanal and commercial fishing also donot have a line of demarcation to separate in the ocean.

The study report from Project Badweyn- Secure Fisheries reveals the composition of the catch in terms of source of catch.

Figure 4: Ratio of domestic to foreign catch of various fish species



A study conducted by Swedish University of Agricultural Sciences, Umea, analysed various data from reports prepared by sources like IOTC, IUCN, FAO, Ogilvy, UNEP and many fishery experts on fish landings/catch between 1950 to 2010 and has come up with an estimated composition of the fish catch in Somalia.

Figure 5: Composition of Fish caught through commercial fishing

Category	Size*	Family/group	Species	Common name	Industrial catch (%)
Pelagic	Large	Scombridae	<i>Thunnus albacares</i>	Yellowfin tuna	7.7
Pelagic	Large	Scombridae	<i>T. obesus</i>	Bigeye tuna	5.9
Pelagic	Large	Istiophoridae	<i>Tetrapturus audax</i>	Striped marlin	0.7
Pelagic	Large	Istiophoridae	<i>Makaira mazara</i>	Indo-Pacific blue marlin	0.3
Pelagic	Large	Xiphiidae	<i>Xiphias gladius</i>	Swordfish	0.4
Pelagic	Large	misc. billfish	-	Other billfish	0.1
Pelagic	Large	misc. pelagic fishes	-	Pelagic fishes	0.9
Pelagic	Small	Clupeidae	<i>Sardinella longiceps</i>	Indian oil sardine	1.9
Pelagic	Small	Clupeidae	<i>Etrumeus teres</i>	Round herring	0.8
Pelagic	Small	Scombridae	<i>Scomber japonicus</i>	Chub mackerel	0.8
Pelagic	Small	Carangidae	<i>Decapterus spp.</i>	Scad	0.3
Pelagic	Small	misc. pelagic fishes	-	Pelagic fishes	0.2
Sub-total pelagic					20.0
Demersal		Lethrinidae	<i>Lethrinus nebulosus</i>	Spangled emperor	11.3
Demersal		Lethrinidae	<i>L. lentjan</i>	Pink ear emperor	5.7
Demersal		Lethrinidae	<i>L. olivaceus</i>	Longfaced emperor	5.7
Demersal		Lethrinidae	-	Misc. emperors	5.7
Demersal		Serranidae	<i>Epinephelus areolatus</i>	Areolate grouper	10.4
Demersal		Serranidae	-	Misc. groupers	2.6
Demersal		Lutjanidae	<i>Etelis spp.</i>	-	4.1
Demersal		Lutjanidae	<i>Aprion spp.</i>	-	4.1
Demersal		Lutjanidae	-	Misc. snappers	2.0
Demersal		Haemulidae	<i>Diagramma pictum</i>	Painted sweetlips	19.0
Demersal		Mullidae	<i>Parupeneus indicus</i>	Indian goatfish	9.4
Sub-total demersal					80.0
Total					100.0

Figure 6: Composition of Sharks and Rays caught in Somalian waters

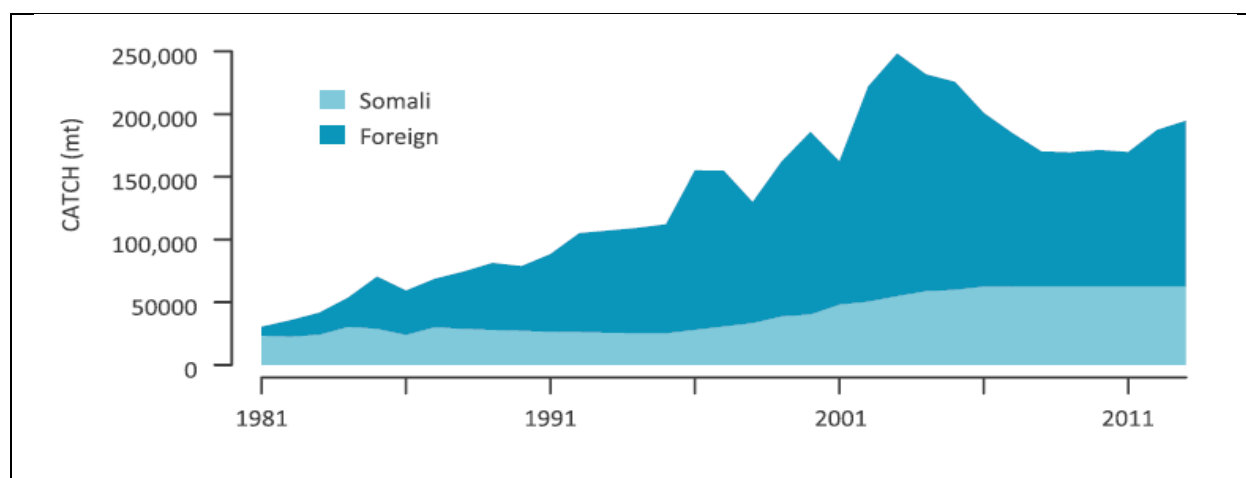
Category	Family	Species	Common name	Catch (%)
Sharks	Carcharhinidae	<i>Carcharhinus melanopterus</i>	Blacktip reef shark	15.0
		<i>C. amblyrhynchos</i>	Grey reef shark	7.5
	Alopiidae	<i>Alopias vulpinus</i>	Thintail thresher	15.0
	Lamnidae	-	Mako sharks	15.0
	Sphyrnidae	-	Hammerhead sharks	15.0
	-	-	Other sharks	7.5
Rays	-	-	Rays and mantas	25.0

Figure 7: Composition of fish species caught (excluding sharks & rays) through artisanal fishing

Category	Family	Species	Common name	Catch (%)
Pelagic	Scombridae	<i>Thunnus albacares</i>	Yellowfin tuna	15.0
		<i>T. tonggol</i>	Longtail tuna	5.0
		<i>Euthynnus affinis</i>	Kawakawa (Little tuna)	5.0
		<i>Scomberomorus commerson</i>	Narrow-barred Spanish mackerel	10.0
	Clupeidae	-	-	5.0
		<i>Sardinella longiceps</i>	Indian oil sardine	4.0
		-	-	1.0
	Carangidae	<i>Selar crumenophthalmus</i>	Bigeye scad	2.0
		<i>Trachurus indicus</i>	Arabian scad (Horse mackerel)	2.0
		-	-	1.0
	Coryphaenidae	<i>Coryphaena hippurus</i>	Common dolphinfish	5.0
	Istiophoridae	<i>Tetrapturus audax</i>	Striped marlin	1.7
		<i>Makaira mazara</i>	Indo-Pacific blue marlin	0.8
		<i>Xiphias gladius</i>	Swordfish	2.0
	Misc. billfish	-	Other billfish	0.5
Sub-total pelagic				60.0
Demersal	Lethrinidae	<i>Lethrinus nebulosus</i>	Spangled emperor	8.0
		<i>L. lentjan</i>	Pink ear emperor	4.0
		<i>L. olivaceus</i>	Longfaced emperor	4.0
		-	Emperors	4.0
	Lutjanidae	<i>Etelis</i> spp.	-	2.0
		<i>Aprion</i> spp.	-	2.0
		-	Snappers	1.0
	Serranidae	<i>Epinephelus areolatus</i>	Areolate grouper	4.0
		-	Groupers	1.0
	Mullidae	<i>Parupeneus indicus</i>	Indian goatfish	2.5
	Misc. marine fish	-	-	7.5
Sub-total demersal				40.0
Total				100.0

The catch landing data in Somalia has to be taken with a pinch since intensive assessments have not been done. There are huge differences in the estimates provided by various agencies. As per the report of **Secure Fisheries Glazer. Et al (2015)**, approximate catch in the year 2013 will be around 200,000 MT, out of which domestic catch will be around 50,000 MT.

Figure 8: Estimated catch by foreign vessels & Somali vessel 1981-2013



Source: **Secure Fisheries Glazer. Et al (2015)**

According to an estimate from SATG group, who has undertaken extensive work in Somalia in the field of fisheries, the shores of Somalia has potential of harvesting 380,000 MT to 500,000 MT of fish annually with the Eastern coast compared to Northern coast share 90% of it.

2.5.Areas/ Regions of Fishing Activity

The artisanal fishing community in Somalia is found across the coast from the top of the country to the bottom. The main fishery areas can be divided into seven main zones, based on major human settlement like: Kismayo, Mogadishu, Eil, Bargal, Bolimog, Las Korey and Berbera. Fisher communities are largely made up of traditional fishermen, living in about 150 fishing villages and towns along the coast from the Kenyan border to Djibouti. Somalia has 65 landing sites in marine waters (MFMR 2017). Catches are landed each day in all 65 landing sites.

Through an FAO project, Somalia has started to registered fishermen and so far a total of 65,144 fishermen have been registered as of 2017. More than half of them are reported to own their fishing boats. The extent of fishing activity region-wise can be analysed from the exercise carried out by the Ministry of Fishery, Somalia to conduct the census on fishing boats. As per the data, collected in 2015-16, the findings are listed below:

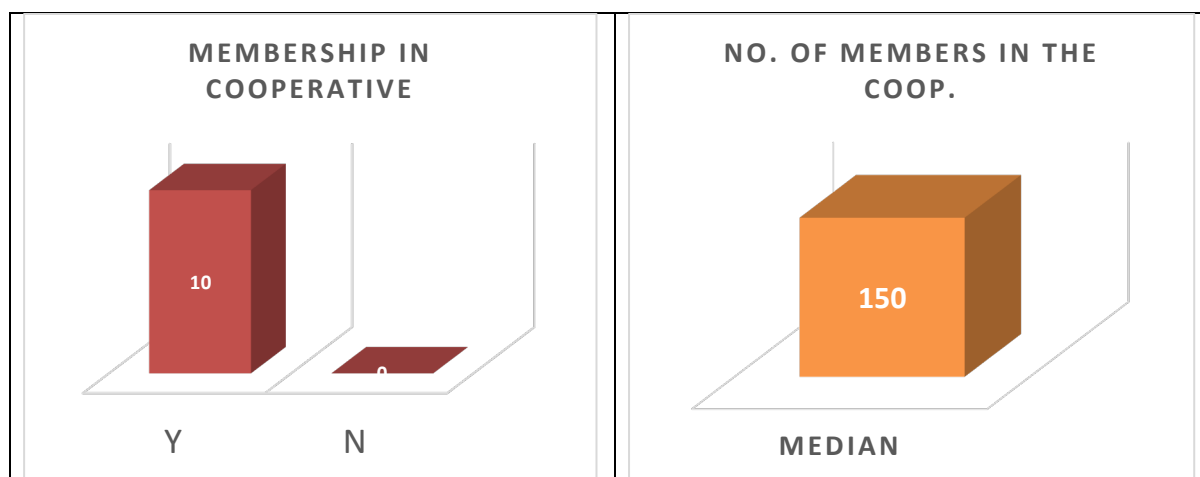
Region	No. of boats	<i>Interestingly, two traditional fishing societies exists in Somalia: The Amarani and Bajuni. Their ancestral roots traces to the Arabian/Persian peninsula and These groups trace their ancestry to Arab and Persian settlers and India and is thought to have landed in Somalia an over 1,000 years ago. The Amarani, are spread throughout fishing communities in and near Brava, Mogadishu, Merca, and inland Afgoye while the Bajuni are found throughout the east African coast from Kismayo to as far south as Lamu, Kenya. These groups look ethnically different from rest of the Somalis and speak Swahili.</i>
Somaliland	594	
Puntland	1697	
Galmudug	565	
Hirshabelle	320	
South-West	386	
Jubaland	495	
Benadir	243	

The above figure provides a snapshot on the prevalence and scale of artisanal fishing in that region.

2.6. Fishery Cooperative

Somalia has seen the cooperative movement since the early 1970s, during Siad Barre's regime. After a series of droughts ravaged the country's grazing lands, the government actively promoted the fishing sector. The Ministry of Fisheries and Marine Transport organized fishing villages into approximately 21 cooperatives. The cooperative structure provided fishing gear and set prices for fish, giving fishers a more stable and viable financial stake. Gradually, due to mismanagement, these cooperative started losing their relevance. In the prolonged civil war, it lost its significance and closed down. Since 2012, many NGOs, FAO, Ministry of Fisheries, Regional governments started aggregating fishermen and promoting cooperatives. It is estimated that all the regions with their fisher settlements have fishery cooperatives. It is estimated around 20 such cooperatives are fully functional and form a federation of fishing cooperatives. Somalia have interestingly a Union of federation of cooperatives called Somalia Union Cooperatives Movement (UDHIS) which has 8 cooperative federations as members.

Figure 9: Fishing Cooperative figures

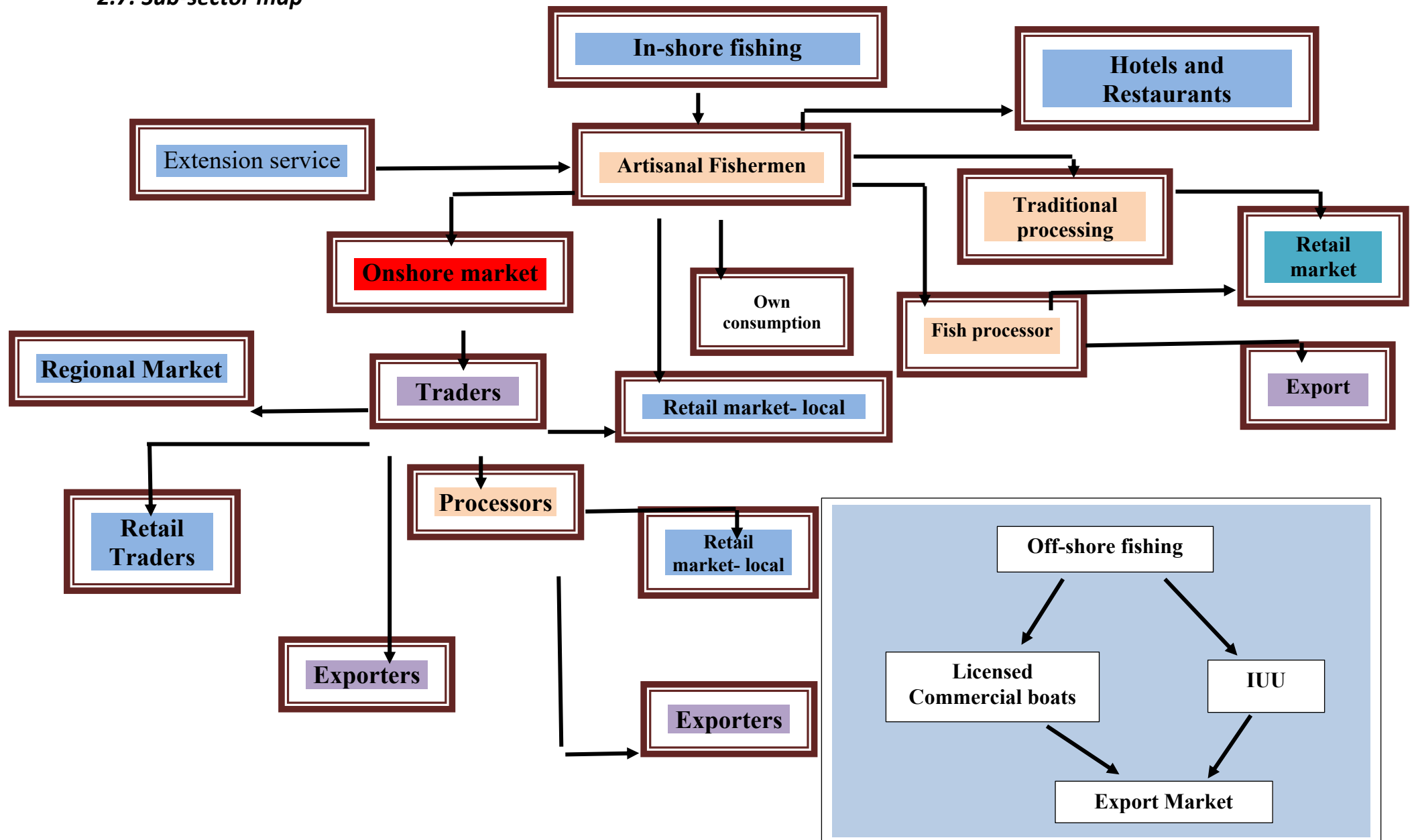


It can be inferred that all the fishermen interviewed as respondents are members of the fishing cooperatives indicating that the cooperatives presence is far out-reaching and can be a major tool for aggregation and value addition. In addition, the size of these cooperatives are big enough to play an influential role in the community.

As per the responses from the respondents; these cooperative plays the following role:

- Training or capacity building of its member fishermen. Maybe, many of the international fishery support projects and NGOs leverage these Cooperatives to provide training inputs.
- The Cooperative supports the fishermen who loses their boats at high sea and help them to retrieve.
- The Cooperative arranges finances for the community members to procure equipment /gears.

2.7. Sub-sector map

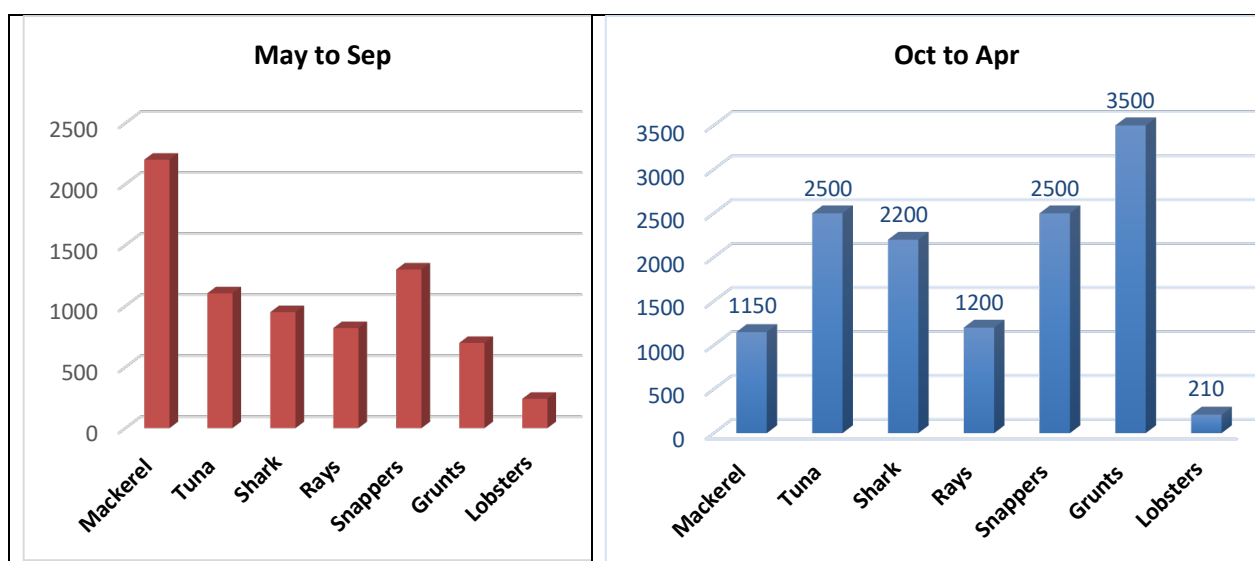


3. FISH PRODUCTION SYSTEM

3.1. Fish catch

The respondents of the study indicated that fishing can be classified into two seasons (**May to Sep & Oct to April**), where the species caught and their volumes varied. The findings presented below indicates the annual median catch (in Kgs) for an individual fisher-person.

Figure 10: Annual fish catch - Artisanal fishermen



From the above chart, it can be easily inferred that Oct to April is the season where the fish catch is the highest. The only exception is the volume of Mackerel caught.

3.2. Fishing Method

Though Somalia has 3200 km long coastline, fishery as industry has not been properly developed with most of the fishery activity can be classified as artisanal with focus on domestic consumption which too is not a very big market due to culinary habits of the population.

Commercial fishery is gradually being promoted. Somalia has granted fishing licenses to 31 Chinese vessels to exploit tuna and tuna-like species off its coast in a bid to tap the sector for economic growth.

Somalia granted fishing licenses to 31 Chinese vessels to exploit The vessels are associated with the China Overseas Fisheries Association, a distant-water trawling group created in 2012 to promote the East Asian giant's competitive fishing edge abroad.

Starting this month, The ships will be allowed to operate for a one-year period in Somali waters, with the agreement stipulating an automatic renewal for an additional year. These foreign fishing vessels will not be allowed within 24 nautical miles from the coast to safeguard the traditional fisher-folk.

The focus on the fishing systems will be based on the artisanal or traditional form of fishery.

A) Boats used

The various type of boats used by the fishermen can be classified into following types:

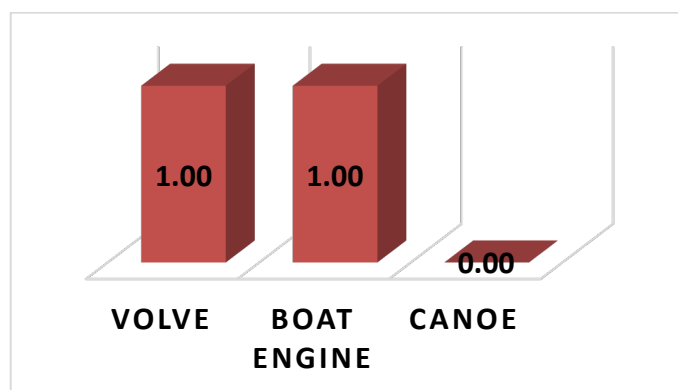
- **Volve-** These are the largest boats used by the fishermen and is approximately 15-20 m in length. It is made of fibre-glass and carries five crew, one captain in the boat and other fishermen to catch fish.
- **Boat engine-** These are boats of 8-12 m in length and also is made of fibre-glass.
- **Canoe (Houris)** – These are the traditional boat used by the fishermen and is made of wood. It is approximately 4-5 m long and carries two crew.



Courtesy: ICRC, VOA

The respondents were assessed on their boat availability; the following information was available.

Figure 11: Per capita boat possession of the fishermen



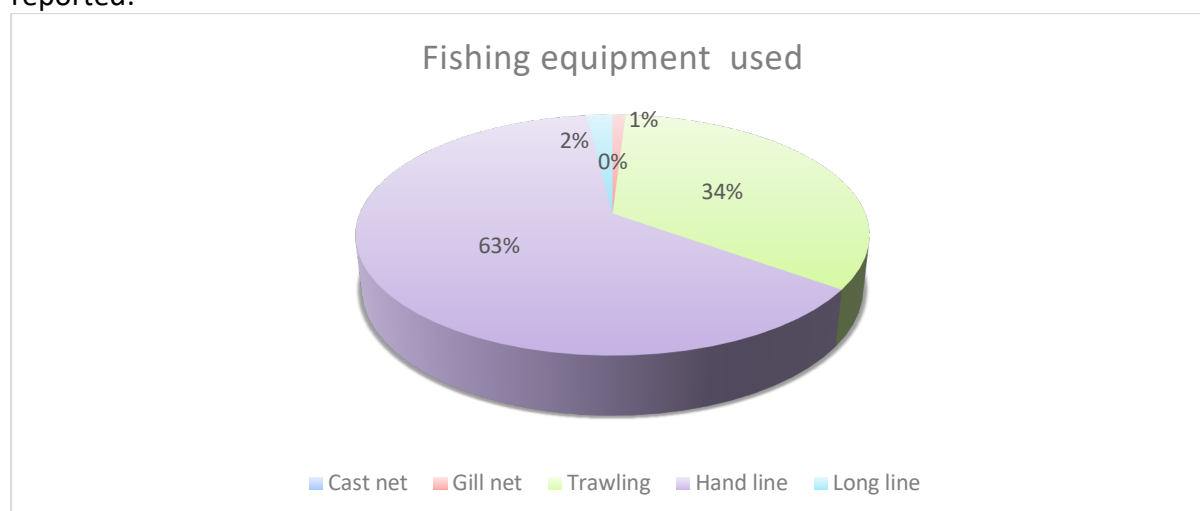
The above chart infers that most of the fishermen are very artisanal with fleets on average includes a Volve and a boat engine.

For the artisanal fleet, the number of fishermen and fishing vessels is largely unknown, however, through an FAO project, Somalia has started to registered fishermen and so far a total of 65,144 fishermen have been registered. More than half are reported to own their fishing boats and 50% are members of fishing cooperatives.

B) Fishing gears/equipment

The fishing gears used by fishermen for catching fish depends on the type of boats used, depth of sea where fishing is done, type of fish to catch. Normally, they use hand lines, gill nets and longlines. Longlines are also used for shark, tuna and other big fish species such as king mackerel, which are the most popular and most favoured species in the country, while Gillnets are used as bottom-set nets for catching shark species.

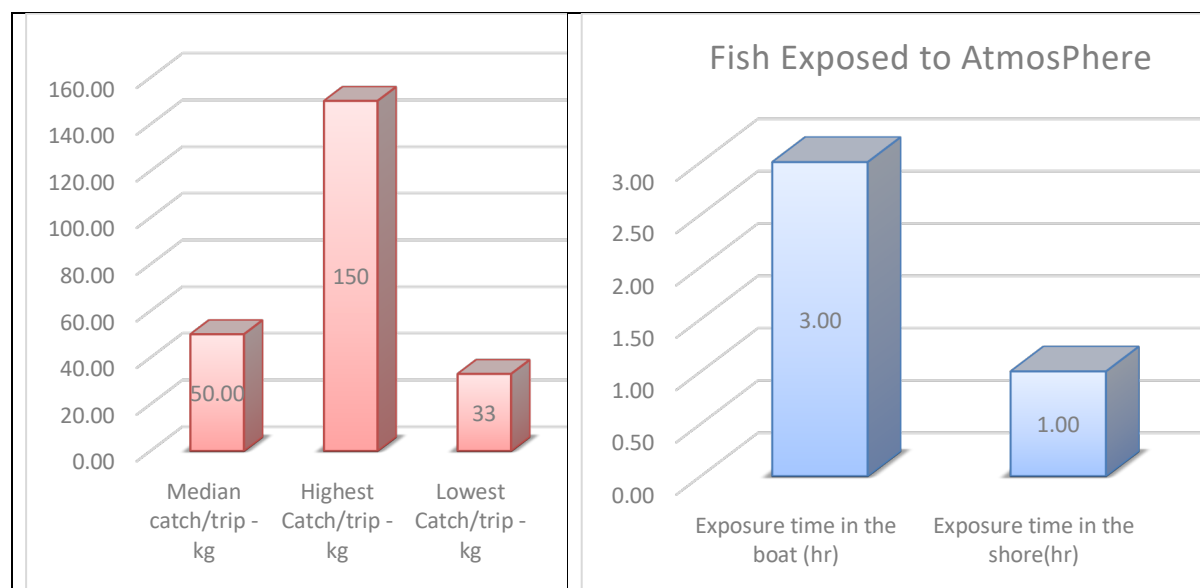
In order to understand the fishing equipment used, the respondents (fishermen) were queried on the various equipment used for fishing in their areas. The following findings are reported:



C) Fishing Schedule

The fishing activity is undertaken six days a week with the fishermen going out to the sea around 4 A.M. and return back to the shore around 10 A.M. The average time the fishermen

gives for catching fishes is around 3 to 4 hours in the trip. Normally, the fishermen venture out to 25 to 30 miles from the shore to catch fish.



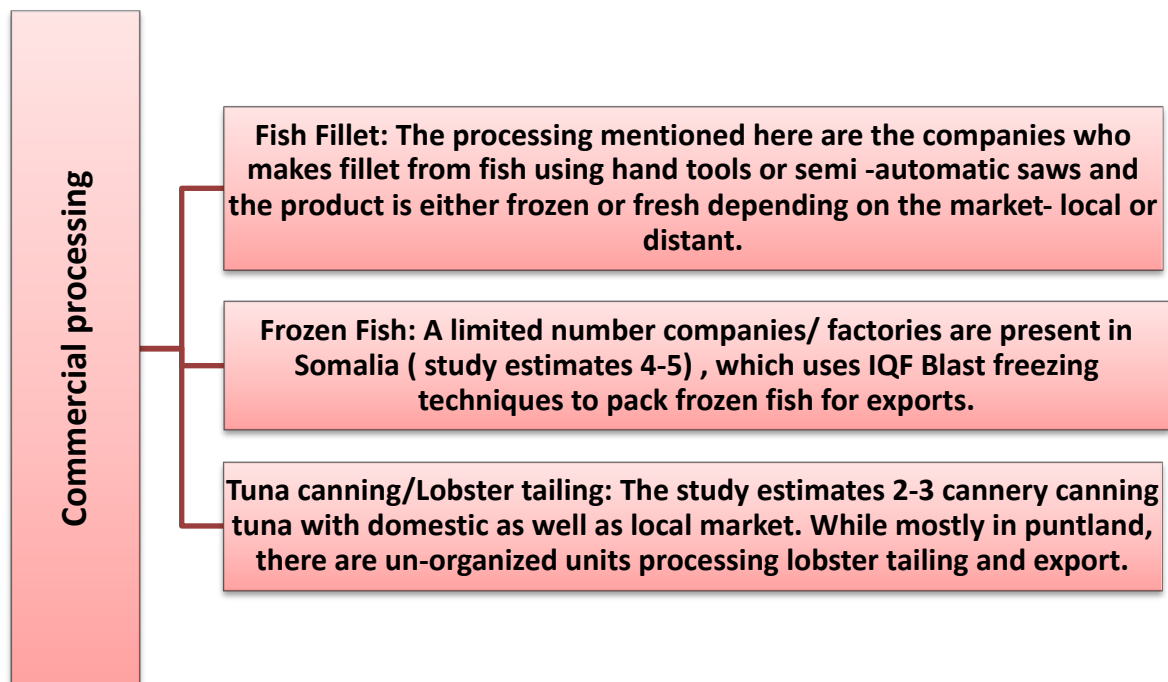
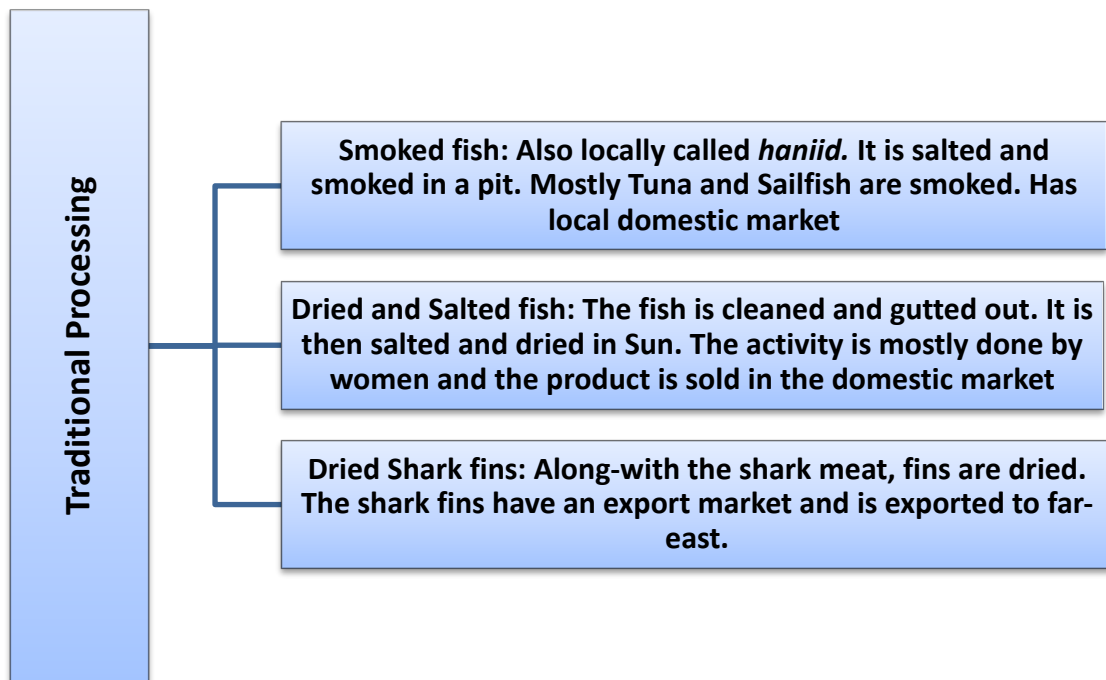
The above table provides an insight into few parameters in the fishing activity. On average 50 kg per boat is exposed to sun for nearly 4 hours. Since, the quality of the fish deteriorates due to excessive and improper handling as well as exposure to sun, it can be seen that around 4 hours of exposure to outside atmosphere is reported. 90% of the respondents (Fishermen) to the study mentioned that they carry ice-box to the sea, which indicates large scale deterioration to the caught fish is minimized.

D) Ice availability and chill room availability

There are few ice-making plants spread mostly in the coastal areas. Though not specifically producing for the fishery sub-sector, their location indicates that the primary customers are the fishermen, traders and exporters. The availability of cold storage facilities is one of the factors that is hampering the fish storage capacities. Some of the ice plants have chill rooms and also some retailers have deep freezers to store. According to Secure Fisheries Glazer et al (2015) report, figures quoted from 2010 show some Puntland settlements had ice plants (in Las Koreh, Bosaso, Bargal, Ras Hafun, Bar Madobe, Bander Beyla, Durdura, Garad, and Dhinowda) and cold storage facilities (in Las Koreh, Bosaso, Habo, and Bargal). These facilities had chill-rooms and some had longer-term deep freezers. In addition, there were over 100 heavy-duty refrigerated trucks and mobile reefer units which were used mainly in the lobster fishery. The exact operational units as of day, is not available to be reported.

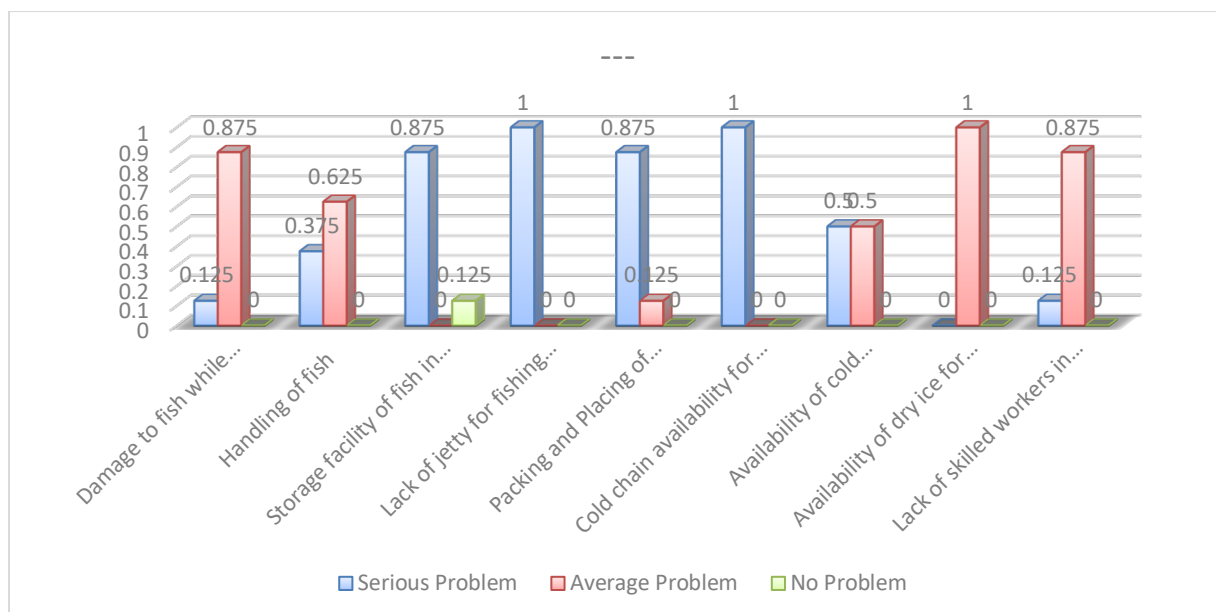
3.3. Fish processing

Fish processing in Somalia can be classified into two broad groups. One is the traditional or artisanal level of processing which basically is done with the objective of preserving fish and to be eaten later. The other type is commercial processing, which is mostly mechanised and targeted to cater to the export market. The number of units of the second type is limited and the study estimates around 10 to 12 such units, with major concentration is in Puntland.



3.4.Problems associated with the fishery supply chain

The study tried to understand the various issues or problems from the other stake-holders in the supply chain namely the traders and processors to the existing fishing production system. The major issues reflected are shown in the figure below:



It can be inferred from the above chart that the perception of the stake-holders in the fishing production system is more related to infra-structure bottlenecks and comparatively less on skilling parameters. This could be due to the existing situation where the fishing activity is still traditional or artisanal without much value addition happening in terms of further processing as well as consumption demand is still marginal.

In addition, interactions with various experts and stake-holders also reveals few more bottlenecks that can play very critical role for further advancement of the fishery sub-sector.

- The existing fishing jetties have infra-structure problems to dock in larger fishing vessels as well as equipment for loading/unloading of Ice storages or transportation of ice boxes out of the jetty. This will be bottle-necks for increased fishing activities as well as adopting modern practices.
- Inadequate and irregular supply of ice-blocks to the fishermen, traders and processors could become a major issue, if the promotion of fishery may lead to increased requirements.
- Almost 2/3rd of the Somalian coast-line is exposed to strong ocean currents and very few natural harbours. This provides limited cover to the fishing vessels during the monsoon period. Kismayo, in Jubaland, is the only natural port; Mogadishu has only a small naturally protected area at the old port, in addition to manmade cargo-oriented commercial facilities

4. FISH MARKETING SYSTEM

4.1. Classification of marketable products

The study reported that the fishery sub-sector in Somalia is pre-dominantly based on trading of whole fish – fresh or frozen. Some processing or value addition is reported but it is very basic and rudimentary. Some mechanised fish processing unit for making fillets, frozen whole fish has come up in Puntland, Juba-land and Banadir regions. Interestingly, some of the secondary study reports has mentioned the existence of canning unit but it has not been reported by our respondents. One unit through secondary sources was identified, which cans tuna fish in Puntland.

Whole Fish- fresh: This the segment which is locally traded in the markets and is focussed towards house-hold consumption, consumption in hotel and restaurant etc.	
Whole Fish- Frozen: This group of products can be further classified into Chilled Fish and Frozen Fish. The chilling of fish is done through packing in ice box and is targeted to be sold at the regional market or neighbouring countries for re-exports. The frozen fish are the ones using modern IQF technique of blast freezing (the secondary research has revealed few factories in Puntland) for exports to countries far off like Oman, UAE etc.	
Fish Fillet- Fresh and frozen: The process of making fish fillet is mostly traditional handmade or semi-mechanized in some commercial factories. The target market is both export and domestic (particularly the hotel and restaurants)	

Cut Shark Fins:

The market is mostly exports. The cut shark fins, most cases the dorsal fins are either dried or frozen.

**Smoked Fish:**

Somali speciality made from tuna and sailfish, which is salted and then smoked in pits and sold locally and called *Haniid*.

**Dried and Salted Fish:**

The fishermen and traders when ice is not available, sun dries after pickling the fish with salt. The dried and salted fish has both domestic and market in Kenya. Mostly Shark meat has the largest demand.

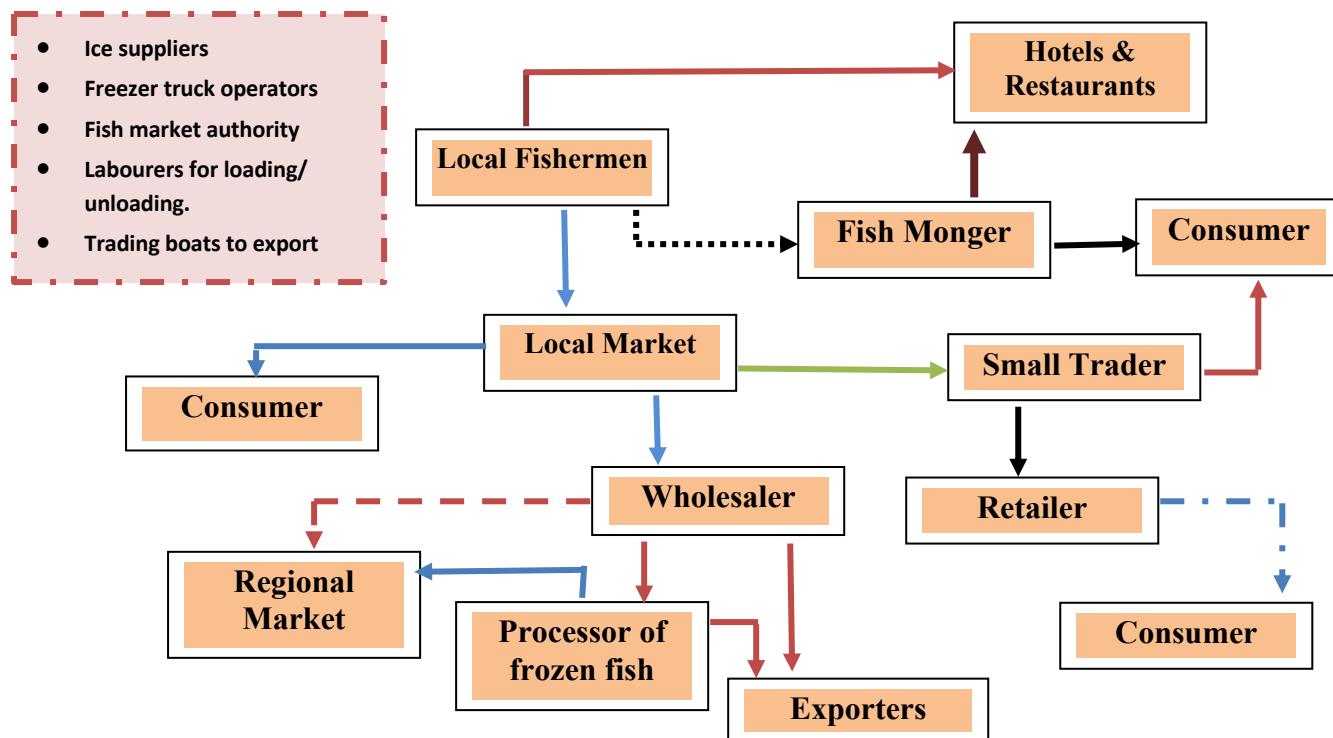
**Lobsters:**

The artisanal fishery fishes Spiny lobsters in the narrow water and sold to traders to be frozen and exported to the neighbouring countries, while commercial fishery catches two types of deep sea lobsters and exports.



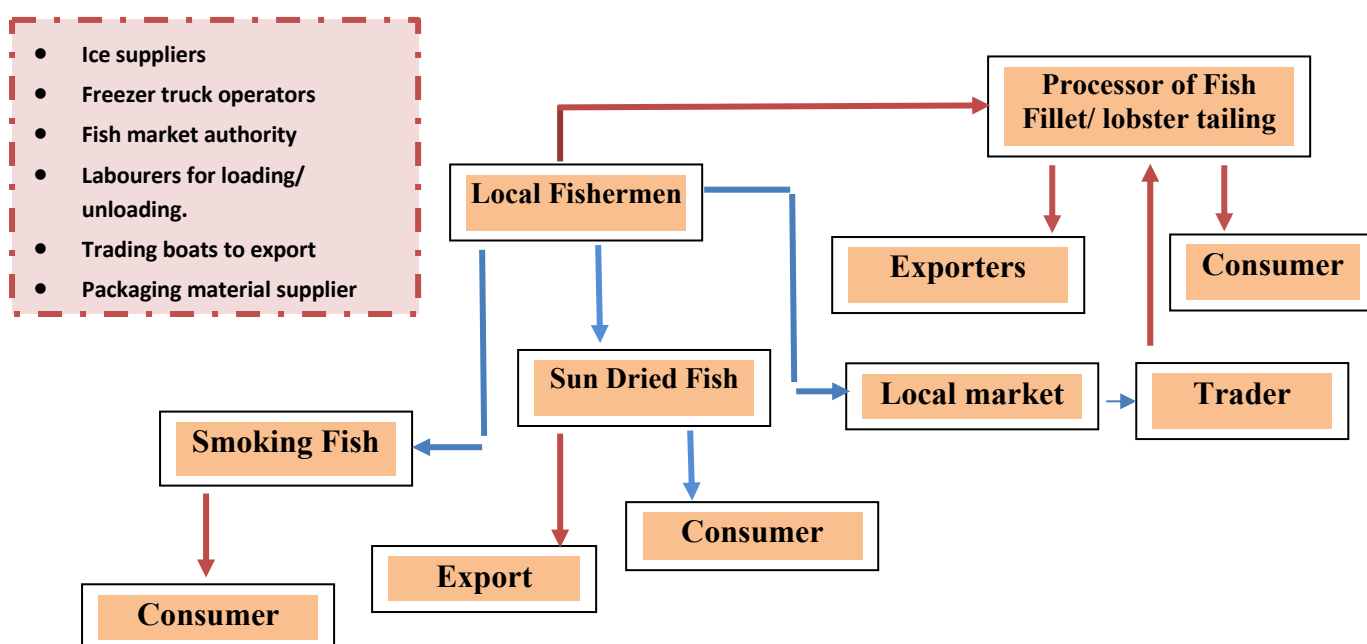
4.2. Various actors in marketing and sales channel of whole fish

The value chain of whole fish trading (both fresh and frozen) in the artisanal fishing category has many sub-chains with various actors playing different roles. A synopsis of these sub-chains is being presented below:



4.3. Various actors in marketing and sales channel of processed fish

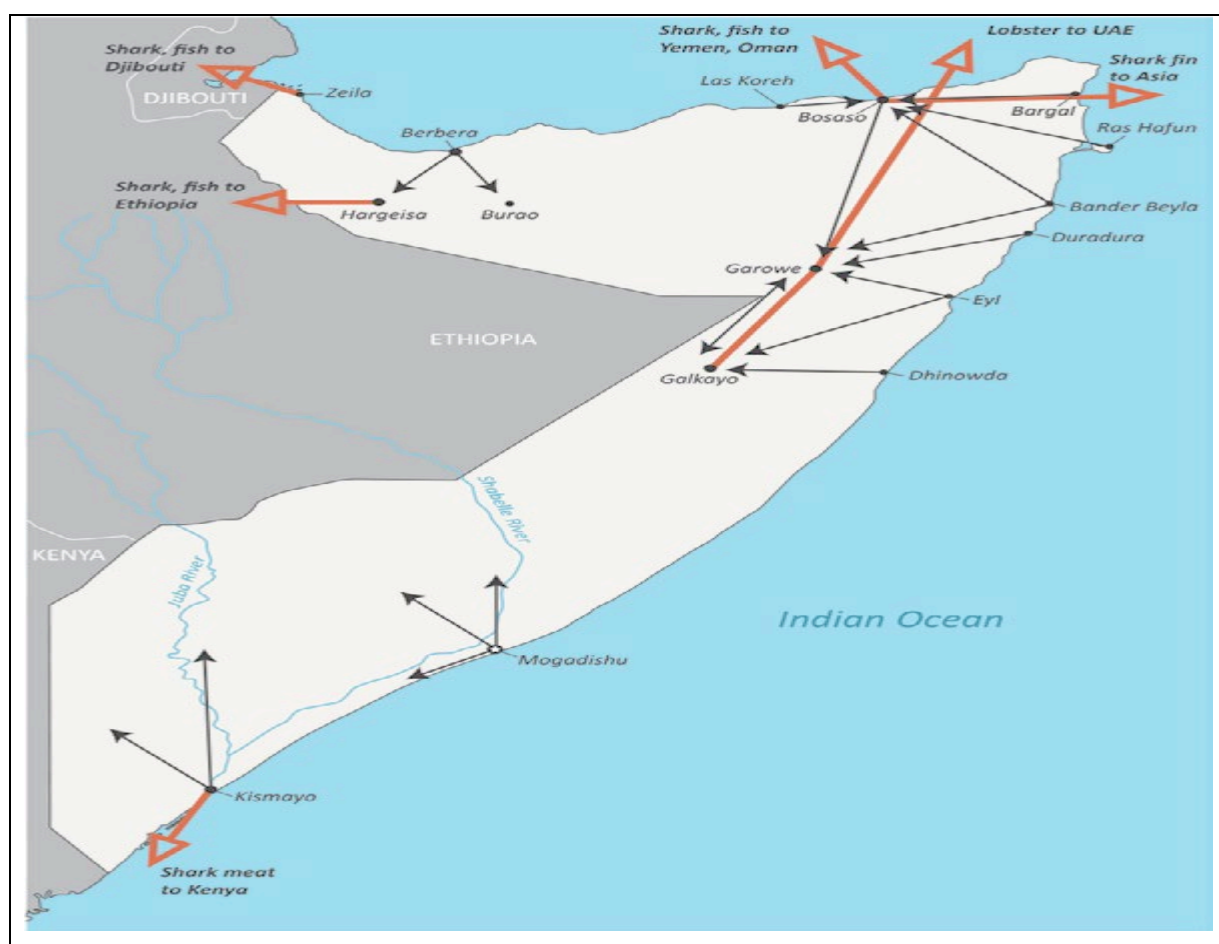
The value chain of processed fish trading in the artisanal fishing category has many sub-chains with various actors playing different roles. A synopsis of these sub-chains is being presented below:



4.4. Fish Trading Route

The sales channel analysed in the previous section indicates that trading of fish and fish products are either for the domestic consumption or the export market. In case of export market, it can be roughly assumed that the channel depends on the end market of the product. The figure below presents a rough snapshot on the trading routes of fish and fish products in the country.

Figure 12: Fishing trade routes in Somalia



Courtesy: SecureFisheries

4.4.1 Domestic Market

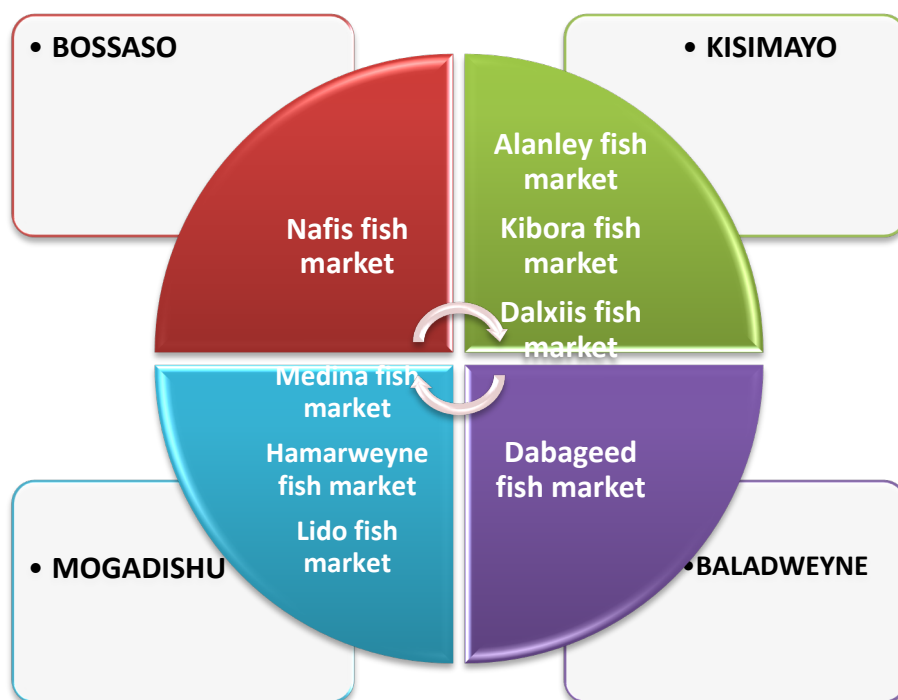
The domestic consumption of fish is very limited in Somalia, and the major consumption centres are in the coast. This is due to many reasons including cultural and also shortage of freezer vehicles to take the fish into the hinterlands. The law and order situation where land transportation to a large extent is weak in Somalia, so even freezer trucks availability is not the immediate solution.

Many of the fishermen, being unable to transport the fresh fish due to lack of ice box and transportation vehicle, convert fresh fish to either smoked fish (*Haniid*) or dried & salted fish.

The smoked fish is mostly for the domestic market while the dried one has markets in the neighbouring country too.

To understand the place where fish as a commodity is exchanged, primarily it is in two places. First of all, it is in the landing sites in the coast, where the fishermen bring the catch auction it to the highest bidder (trader). The larger transaction takes place in designated markets within the city/town limits where both wholesale and retail trading takes place. The report provides below the major market area for fish trade in the major towns of Somalia.

Figure 13: Major fish markets in Somalia



The figure below provides an understanding of the price and cost of fish trading.

Figure 14: Selling price of the fishermen (USD/kg)

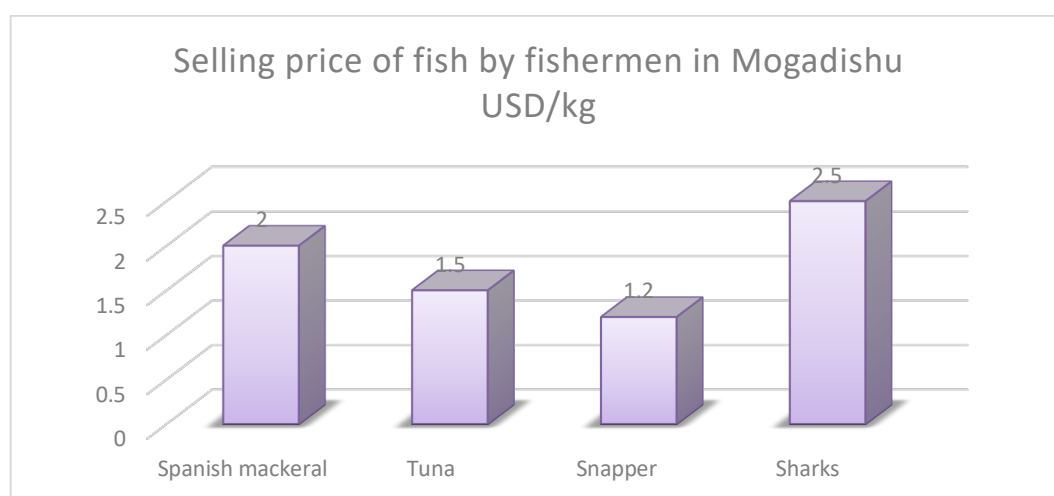
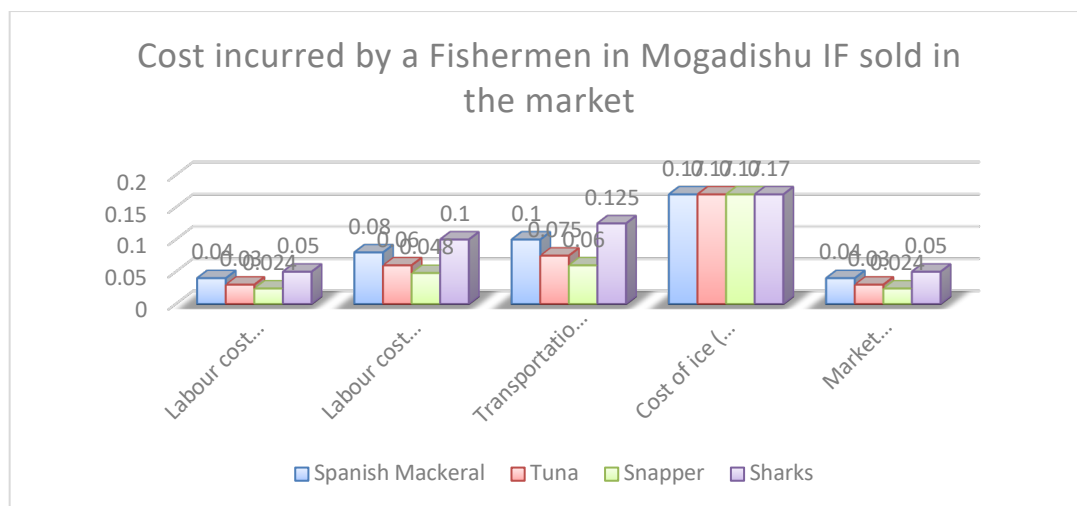


Figure 15: Cost reported by fishermen to trade in the market in Mogadishu



The figure below presents the various costs that a fish trader has to incur.

Figure 16: Prices reported by traders

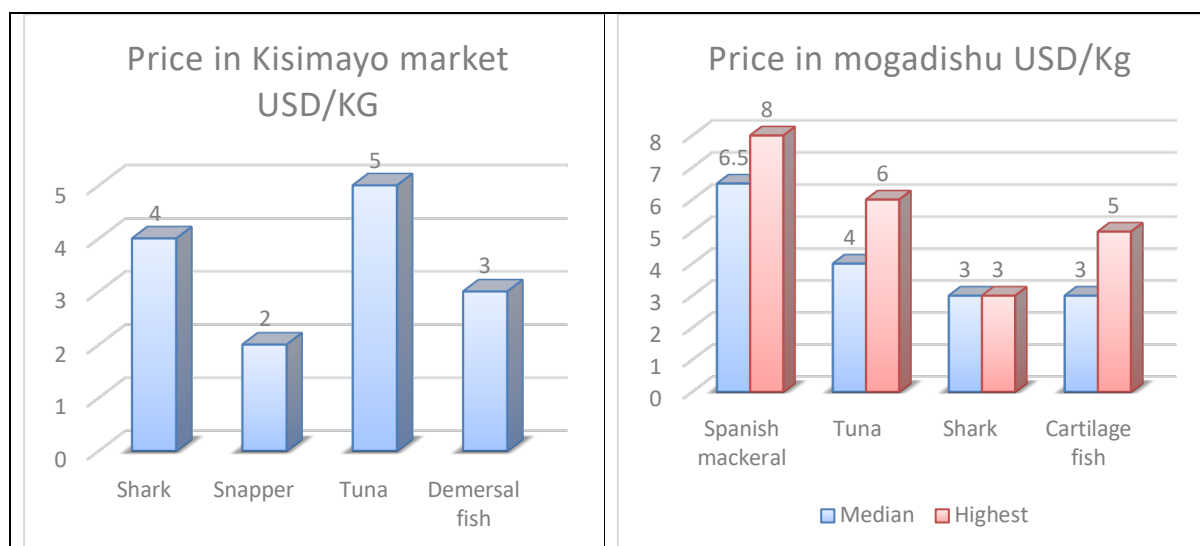
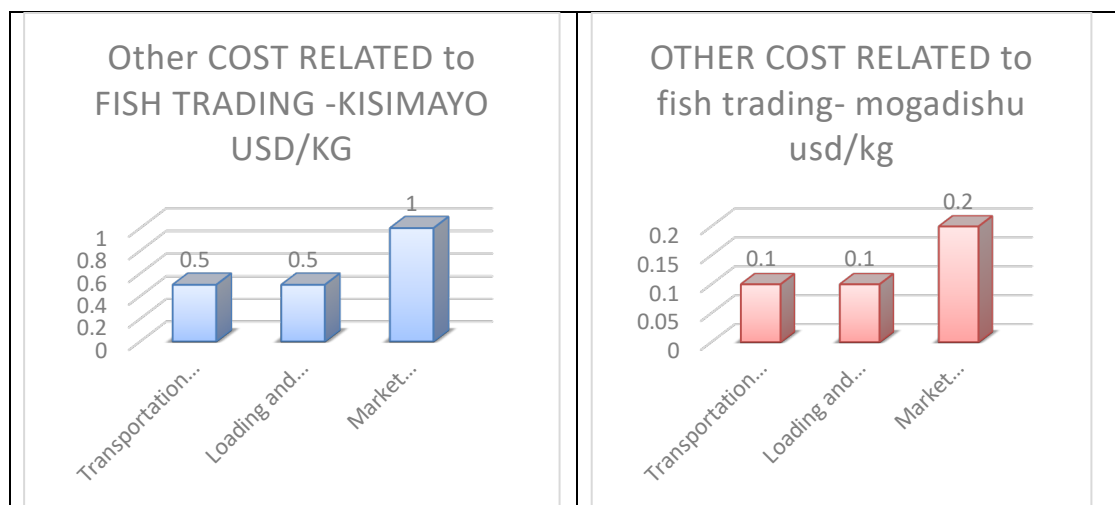
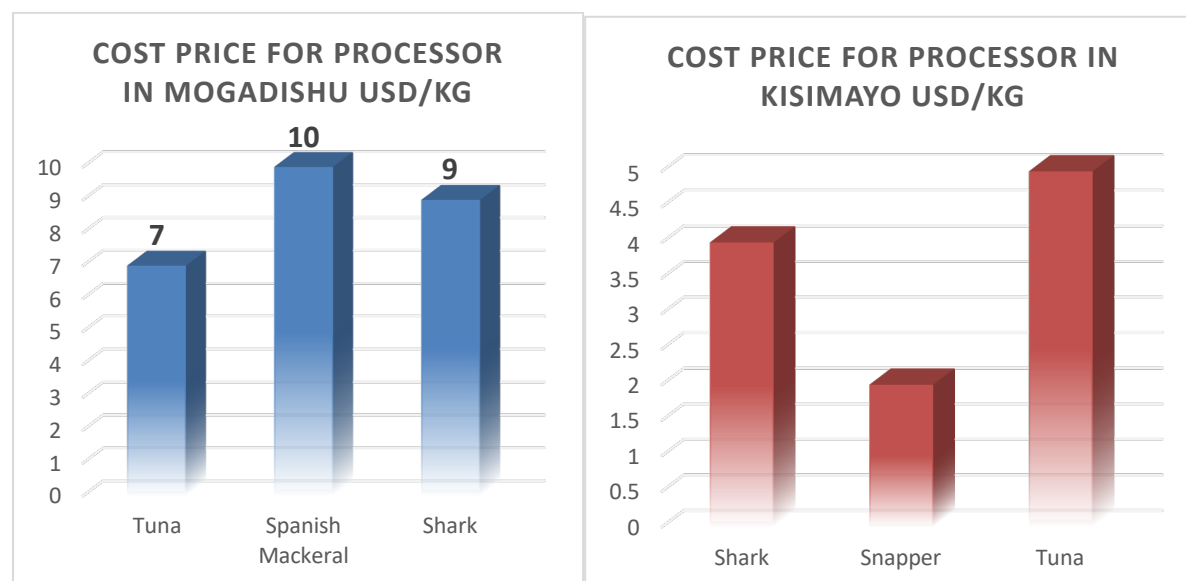


Figure 17: Other cost related to fishing reported by traders



The price reported by the processing units in both Mogadishu and Kisimayo

Figure 18: Reported cost price by the processors



4.4.2 Export Market

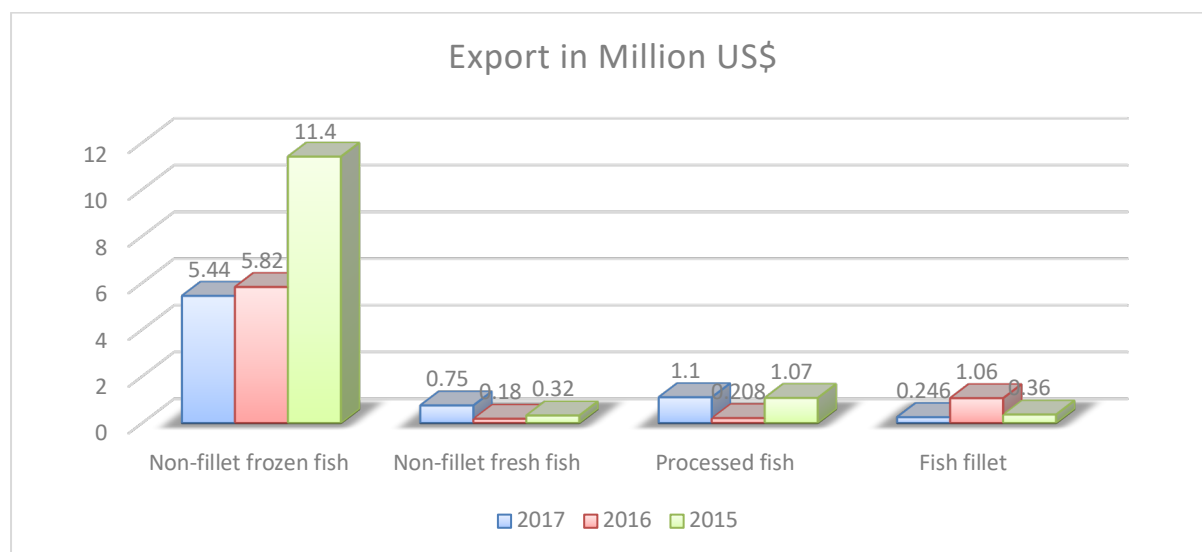
The actors involved in the export can be classified into organized registered exporters which includes traders as well as processing units which are either canning, blast freezing of fish, frozen fillet etc. while the second category are the ones who are either Somali citizen having Yemeni boats or small traders in the South Somalia. These un-organized group transfers fresh fish, dried & salted fish, shark fins, lobster & lobster tails to Yemen, Kenya and Ethiopia. These are either consumed in the local market or re-exported.

The export figures for fish and fish products on Somalia has been taken from the website of “**Observatory of Economic Complexity (OEC)**” and were analysed to find the following findings.

The exports can be classified into four groups:

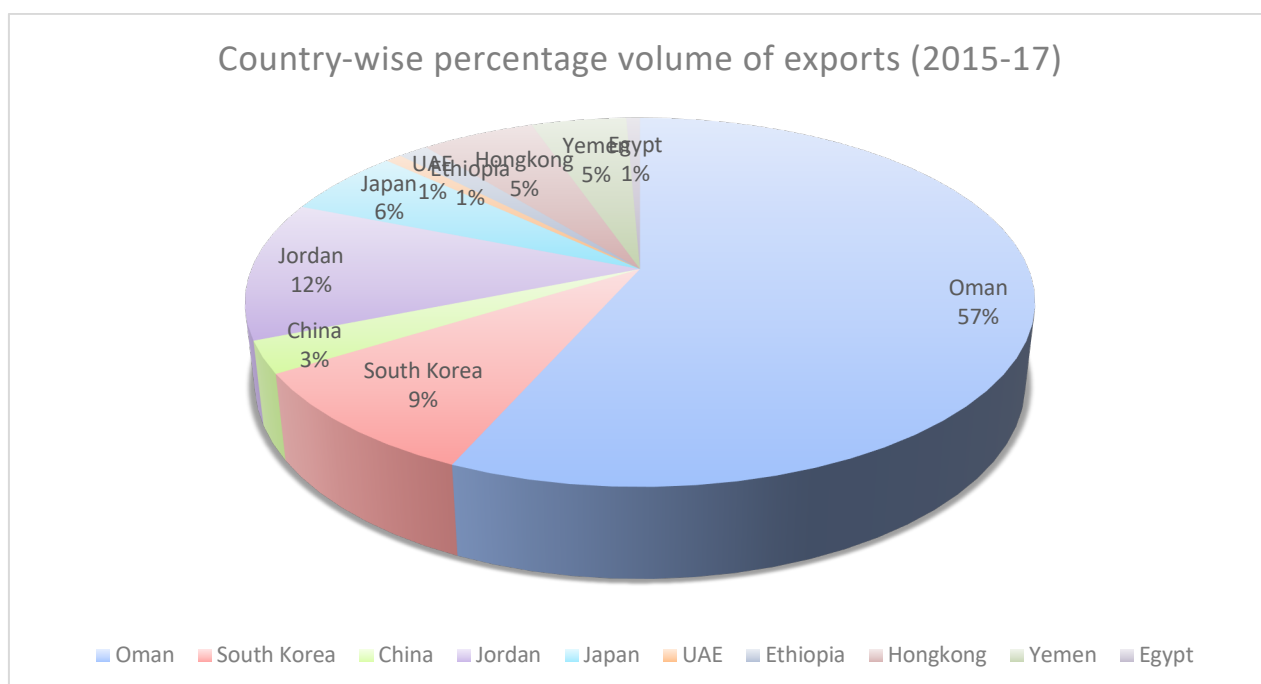
- Non-fillet frozen fish
- Non-fillet fresh fish
- Processed fish (includes- dried & salted, shark fins, lobster tails, canned fish)
- Fish fillet

Figure 19: Export figure of fish and fish products- Somalia



Courtesy: OEC

Figure 20: Export destinations for Somalian fish and fish products

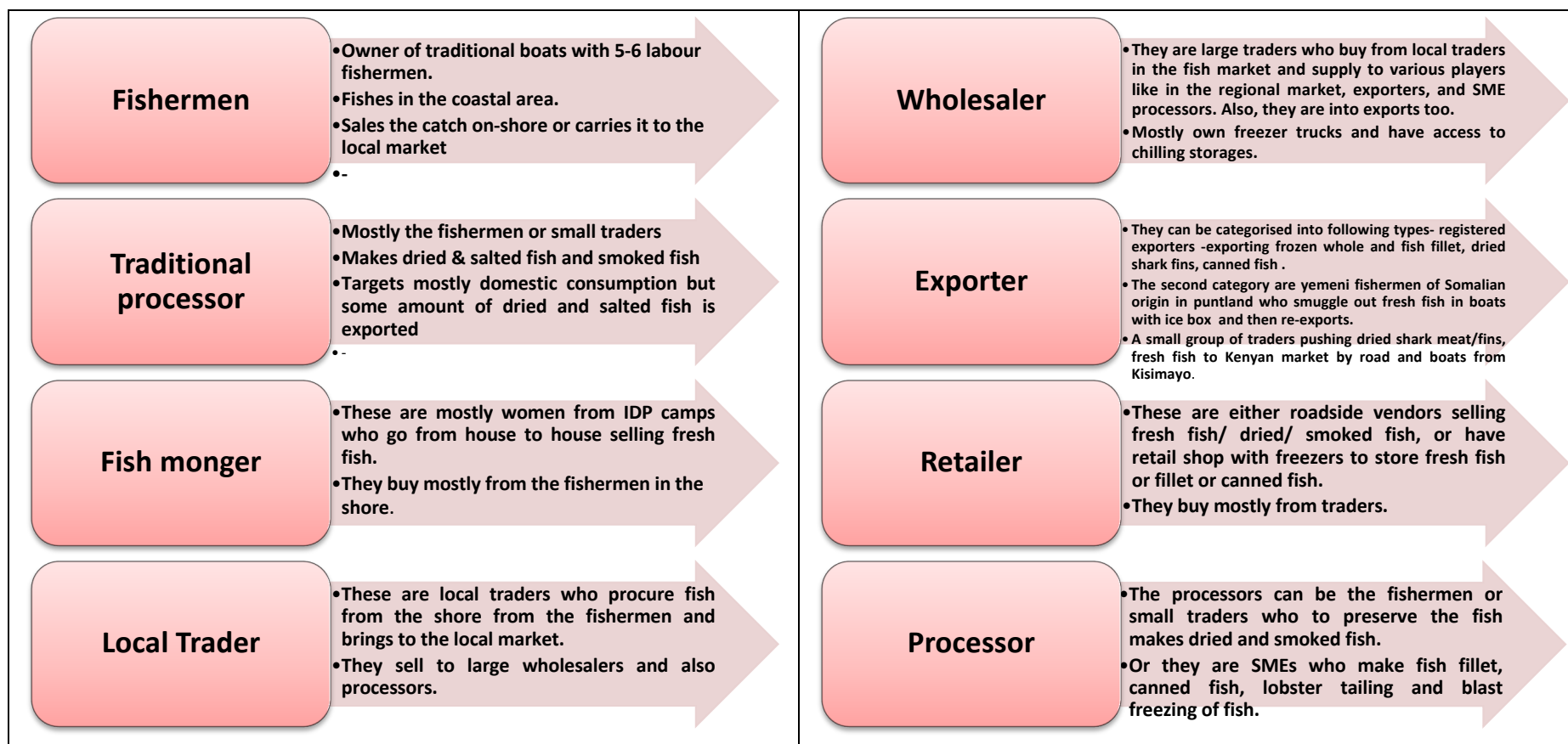


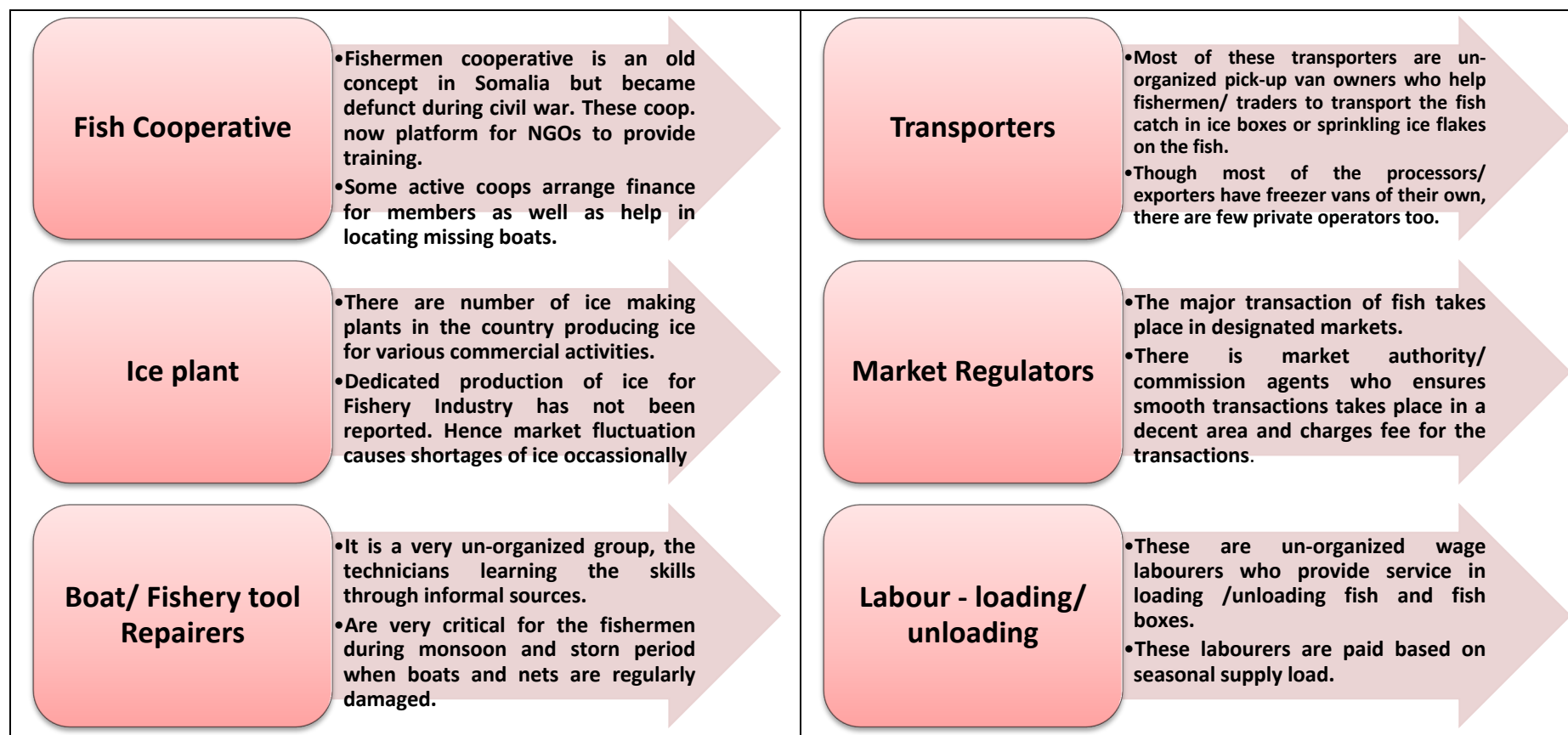
Courtesy: OEC

It can be inferred from the above two figures that the value added products for exports is still in the infancy stage with focus of exports is whole fish. It can lead to another conclusion from the fig. 17, that maybe many of the whole fish is going to neighbouring countries or the gulf countries where produce are processed and re-exported.

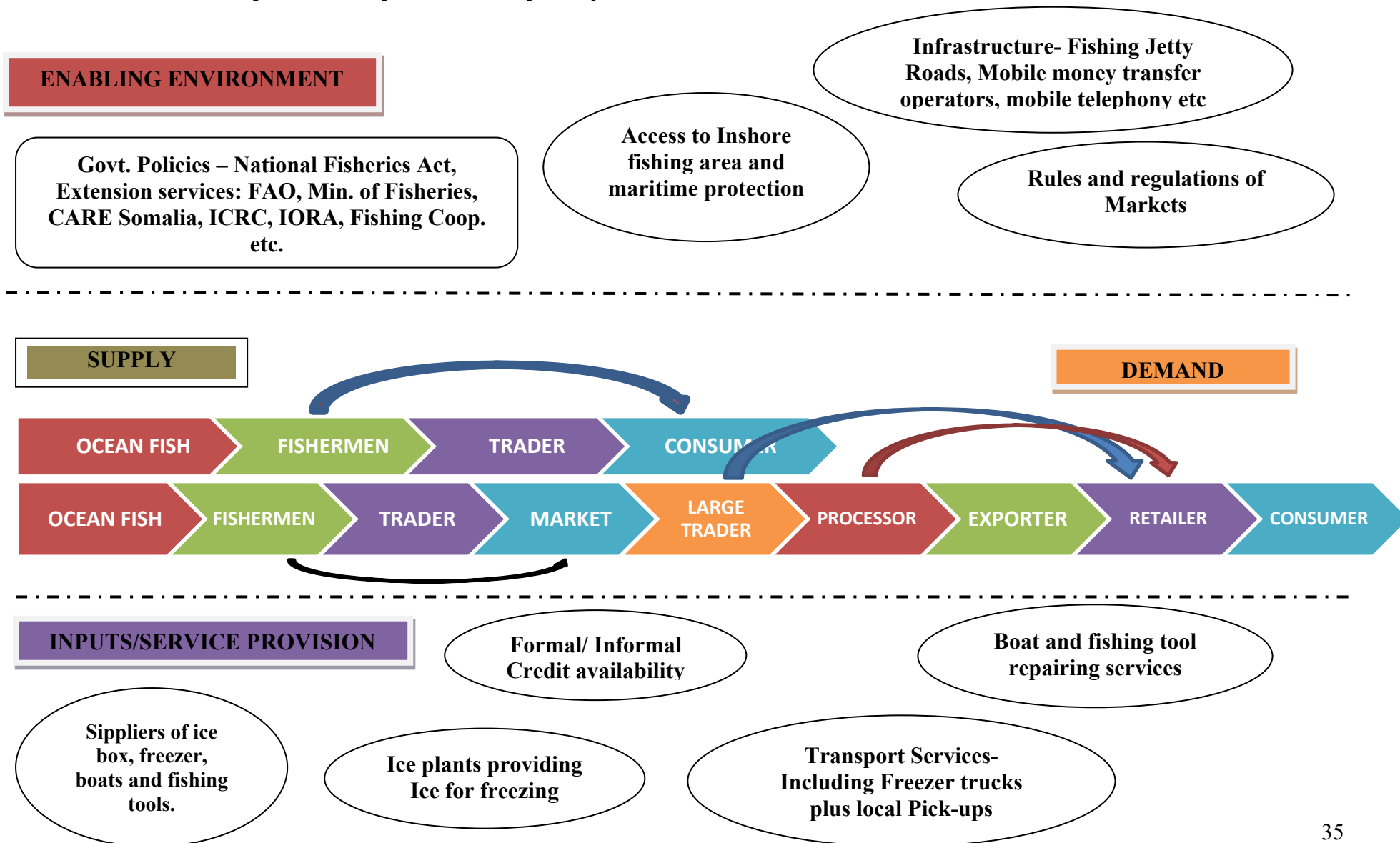
5. VALUE CHAIN ANALYSIS

5.1. Various actors and sub-actors

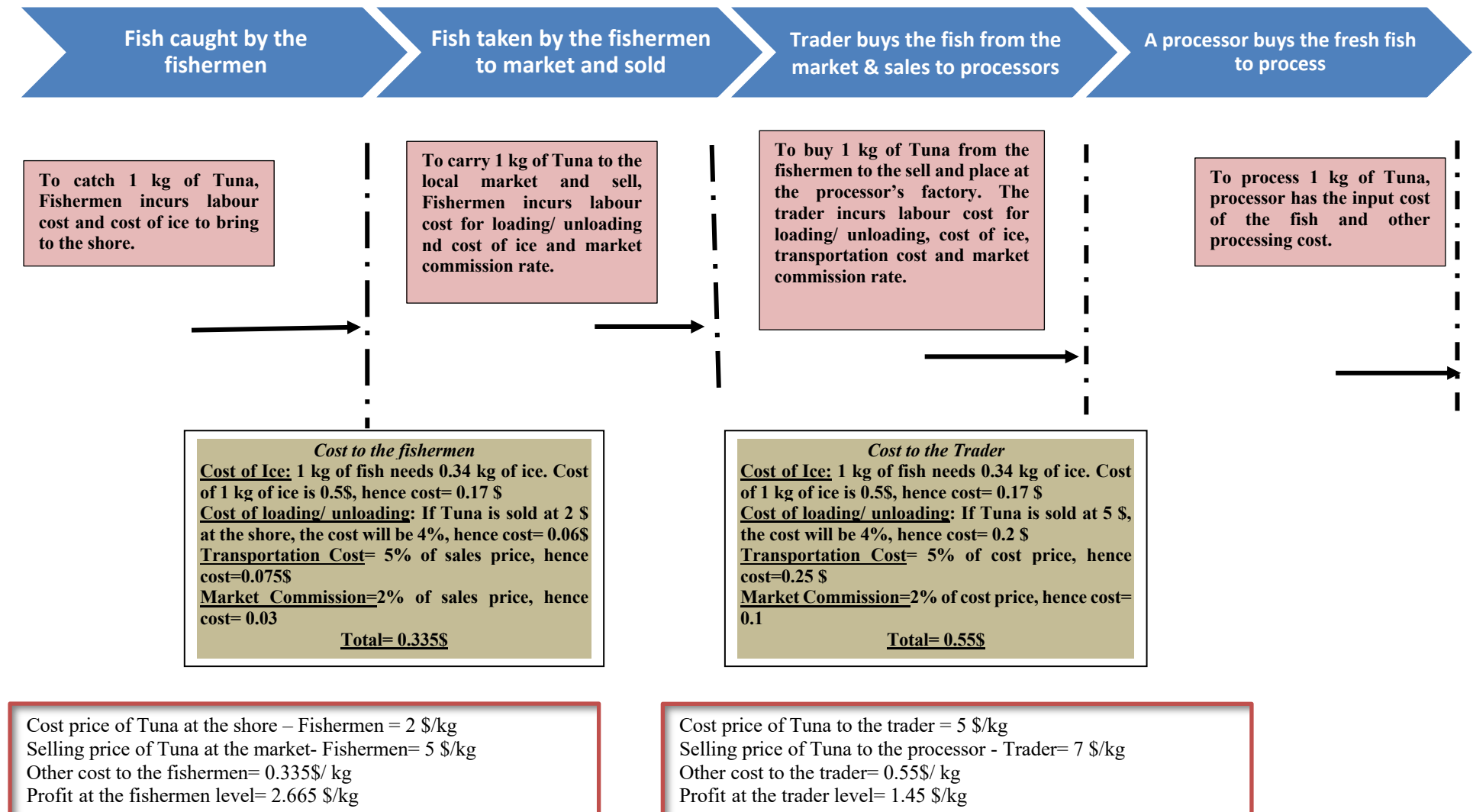




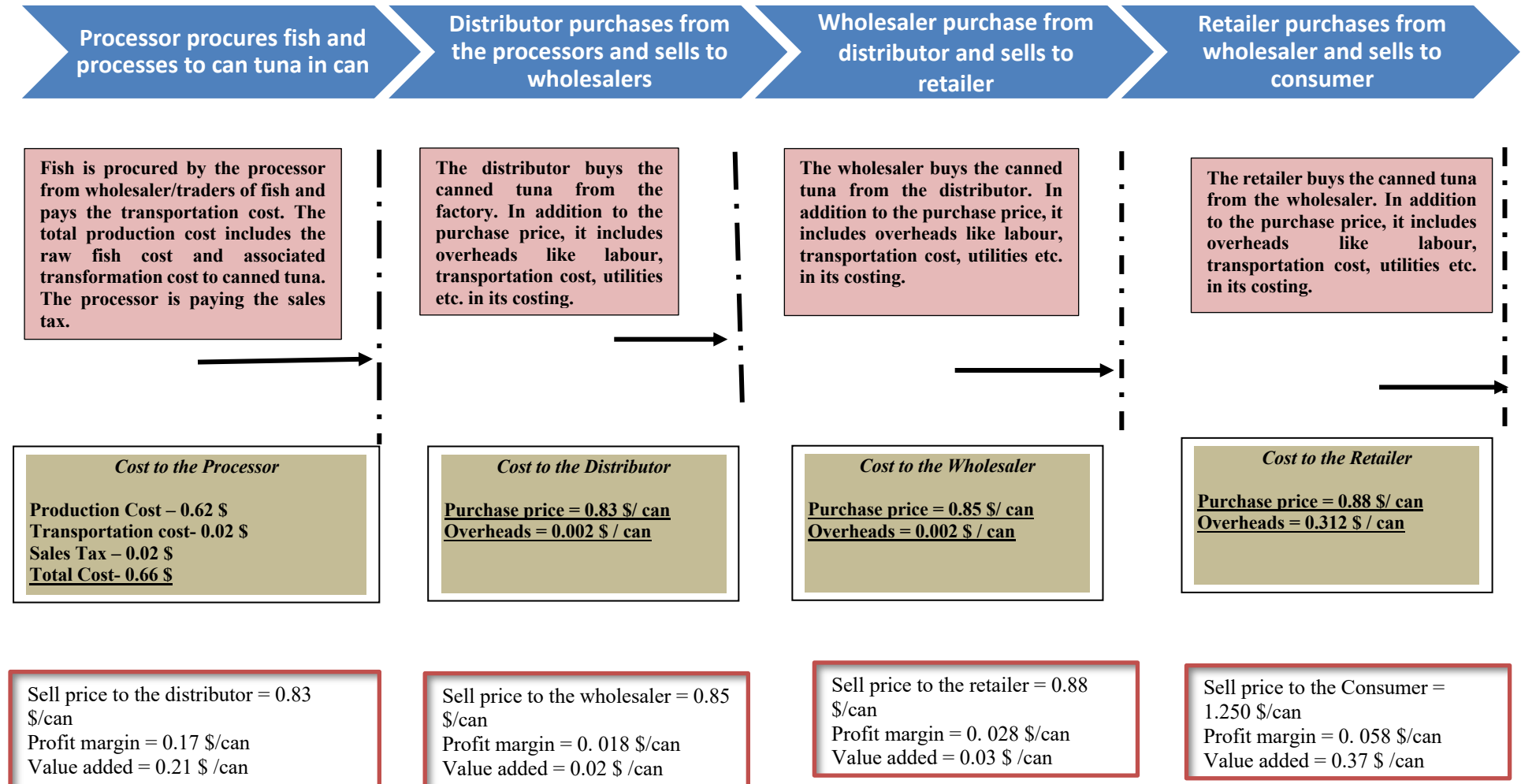
5.2 Value Chain framework for Artisanal fishery value chain



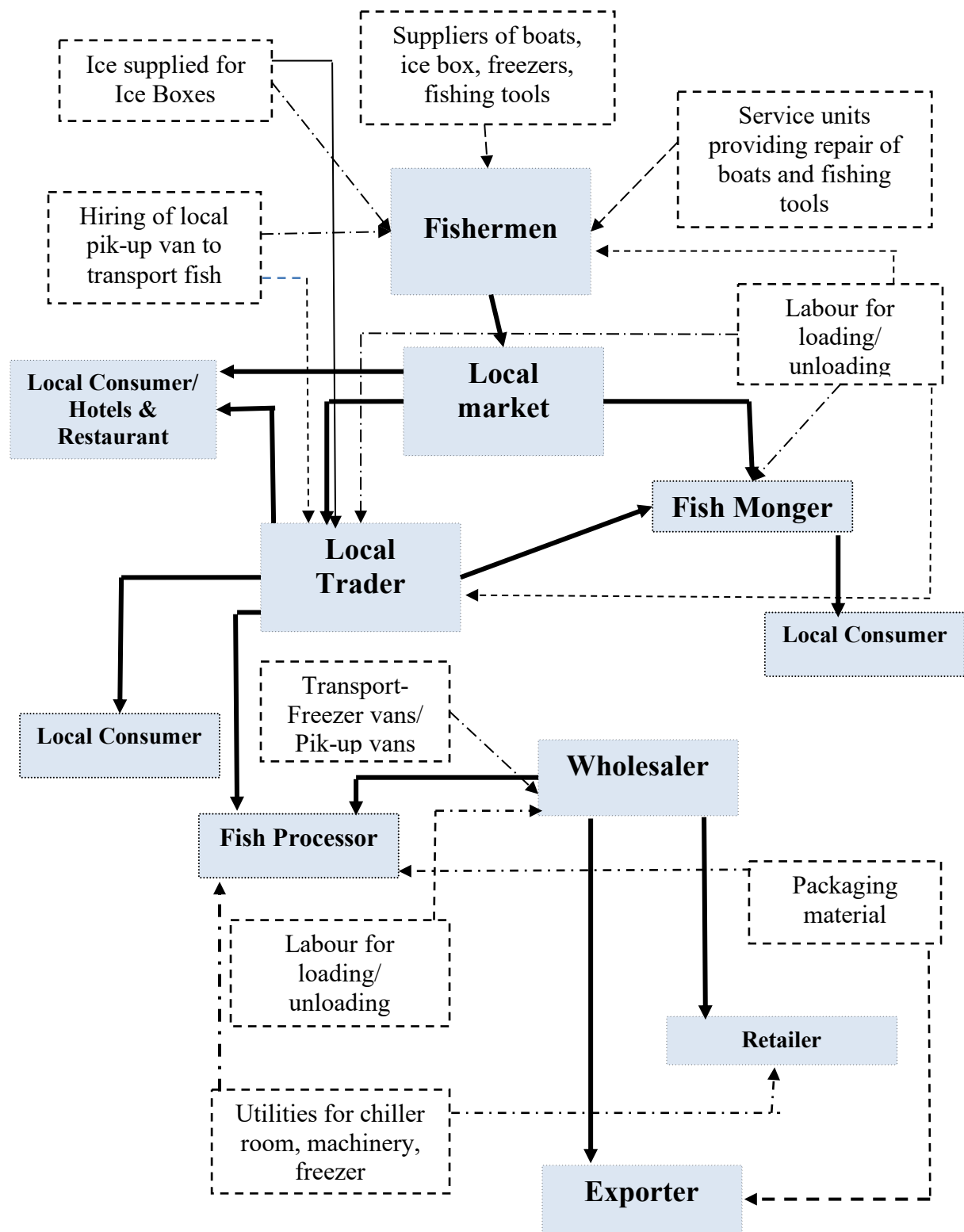
Value addition in Fishing trading value chain (Calculated on primary data)



Value addition in fishery processing value chain- canned fish (Using secondary data sources)



5.3 Supply Chain in Fish value chain



5.4 Chain Governance

Value chains involve regular interactions between the various actors implying repetitiveness in their relationships. Governance ensures that interactions between firms along a value chain exhibit some reflection of order rather than being simply random. Value chains are governed when parameters requiring product, process, and logistic qualification are set which have consequences up or down the value chain encircling package of activities, actors, roles, and functions.

Governance can be categorised into the following types; **legislature** (making the rules), the **executive** (implementing the rules) and the **judiciary** (monitoring the conformance to rules). This governance can be exercised by parties internal to chain or external to chain.

As a part of the value chain analysis for **the fishery sub-sector in Somalia**, the chain governance was looked into;

	Exercised by parties internal to chain	Exercised by parties external to chain
Legislative governance	- The quality of the fish supplied by the fishermen are deciding factor for the prices governed by the traders who purchase from them. - Similarly, the processors requirement decides the pricing for the traders.	- Market committee fixes the commission to be collected for the transactions taking place in the market. - Demand by consumers for Tuna or Shark fins prompts preference in procurement of these items. - Artisanal fishermen are not highly regulated by the authorities. - Authorities do not allow the commercial fishermen to enter the territories allotted to artisanal fishermen.
Executive governance	The fishermen/ traders themselves sort the fishes according to buyer needs before sales.	- Market committee ensures that the commission on sales are paid on time by the traders/ fishermen. - The maritime guards enforce the rules for commercial fishery including reporting the catch and avoidance of the territorial waters dedicated to artisanal fishermen.
Judicial governance	- The quality/ produce conformation is checked by the traders while procuring them from the fishermen. - Similarly, the fish processor checks when procured for the traders.	- Market committee monitors whether the fishermen / local traders are paid according to the agreements. - Fish and Fish products exported legally needs to conform to consumer standards of the respective countries.

5.5 Inter-firm cooperation

Inter firm cooperation is defined as joint activities undertaken by the various stake-holders in the value chain. Cooperation can be subdivided in *horizontal linkages*, i.e. cooperation among similar types of enterprises, for example fishermen, and *vertical linkages*, cooperation between enterprises at different positions in the value chains, for example the local traders and fishermen.

Horizontal linkages refer thus in particular to the regular contacts and relationships between peer enterprises within a specified area, more specifically, it refers to joint activities of such firms, i.e. joint purchases, joint usage of equipment, joint contracts and joint learning

On the one hand, vertical linkages refer thus to contracting and subcontracting relationships between enterprises of different types in terms of scale.

5.5.1 Horizontal linkages

Horizontal linkages, or the cooperation among similar types of stake-holders, are relatively weak. In case of fishermen, the only perceived cooperation is being member of the local fishery cooperative where joint training programs are conducted by some NGOs. Another area of support received through the cooperatives is assistance to the fishermen to locate boats during storms.

5.5.2 Vertical linkages

In general terms, stake-holders in the value chain have strong vertical linkages. The flow of fish to the consumer either whole or processed goes through various stake-holders. The trader will buy from the fishermen, the wholesaler buys from the trader and sells to retailers or the exporters. The processors depending on the location and quantity buys from wholesalers or the traders. All the major actors in the value chain are vertically linked. The wholesaler or the large traders have prior contract of supply to the processors or the exporters. While many hotels/restaurants have informal agreement of supply with the fishermen.

6 FINDINGS & KEY ISSUES

The artisanal fishermen in Somalia is still practicing very traditional form of fishery with obsolete tools and equipments. Most of the fishing boats are still old boats used traditionally requiring frequent repairing. The introduction of new modern boats will also simultaneously require building the capacities of the existing micro-entrepreneurs involved in repairing boats.

Since, most of the fish catch is targeted towards domestic consumption, not much importance is given towards handling of fish, placing the fish in ice box, packaging of fish etc. The existing skilled work-force doing these activities are either not informed or lack the capacity & knowledge to do those activities as best practices. These skill sets will play a very important role, if the fishery sub-sector is promoted extensively, and investment towards export market and processing materializes.

Traditional fish processing like sun-drying and smoked fish is prevalent since the fishing community required to preserve and extend the shelf-life of caught fishes. With growing demand of dried fish in the neighbouring countries as well as distant markets especially for products like dried shark fins & lobster tails, a need for standardization of the processes and efficient quality norms to be built in. In addition, fish drying also promotes women empowerment since the activity is mostly done by women-folk in the community. Standard operating procedures and capacity building will improve the situation.

After the end of the civil war, sensing the huge business opportunity in the fishery sub-sector, few private sector has invested and established mechanized fish processing units. These units are primarily in increasing shelf-life of fish for export through IQF blast freezing techniques, fish canning units, and semi-mechanised or manual fish fillet making. In the coming years, more of these units will be established and it is expected that there will be shortage of skilled work-force in these areas. With that perspective in mind, steps should be taken towards building capacities of Somalian work-force to manage these mechanized units.

7 RECOMMENDATION ON AREAS for SKILL SET UPGRADATION

A) FISH PRODUCTION

Extension Services	
Areas of Skilling	Remarks
Techniques on improving the capacities of fish boat repairers to provide services to boat men.	These training modules will be targeted at the micro entrepreneurs servicing the fishing boats.
Packaging, Storage and Transportation	
Areas of Skilling	Remarks
Technique on how to store fish in ice box, deep freeze, ice chiller to prevent damage in fish due to inappropriate cooling.	These training modules will be targeted at the workers working with fishermen / traders as well as fish processing units.
Handling, stacking of fish, loading, unloading to minimize damage.	

B) PROCESSING

Traditional artisanal processing	
Areas of Skilling	Remarks
Technique of fish drying in Sun: • Removing of guts and cleaning of fish. • Process of Sun drying • Packaging of sun dried fish	This training module is to be designed for community of fisherwomen who is already engaged in similar activities.
Mechanized Processing	
Capacity building in the vocational training spectrum for mechanized processing: • Use of machine to make fish fillet: • Storage of fish in chillers: ○ Placing ○ Temp. range based on species ○ Period of storage ○ Regular monitoring • IQF Blast freezing technique • Packaging for export destination. • Mechanized process of fish canning	Gradually, units will be coming up in Somalia, in-fact few of them already in existence in Puntland and Banadir, which will start mechanized fillet making, IQF blast freezing technique, canning etc. This training module will be geared towards creating a skilled work-force that can be of high demand in coming years.

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